Section I

Introduction and Overview

Chapter 1

Overview and Evolution of Silicon Wafer Cleaning Technology*

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1.1 INTRODUCTION

The preparation of ultraclean Si (silicon) surfaces by cleaning and surface conditioning for the manufacturing of integrated circuits (ICs) has undergone considerable changes since we published the first edition of this handbook in 1993. The driving forces for these changes have been the ever-increasing requirements for producing advanced Si devices with improved performance, reliability, and cost. The feature sizes of these circuits have been scaled down to below 10 nm and the device structure may incorporate multilevel metallization layers with Cu (copper) and ruthenium (Ru), cobalt (Co), and other special dielectric materials, such as low-k (dielectric constant), and compound semiconductors such as SiGe and III-V epitaxial layers. Furthermore, since the second edition, the changes in devices structures have been phenomenal; FinFET (fin-shaped field effect transistors) are used for logic devices and 3D NAND-stacked memory devices and the use of through silicon vias.

The earlier Si wafer cleaning processes have evolved from aqueous-chemical (the so-called “wet chemical”) to alternative dry cleaning processes for metallized device wafers. These processes include gas-phase procedures, such as plasma reactions for stripping organic photoresist masking layers and their residues. The application of cryogenic aerosols and supercritical fluids for removing particulate contaminants for applications where aqueous chemistry cannot be used is still being investigated and in limited production use.

The understanding of surface contamination and defectivity and the role of particle adhesion, deposition, measurement, and removal have been substantially advanced during the past 14 years. The same is true for the ancillary analytical and control aspects, including the elucidation of the chemical composition of Si and SiO_2 surfaces, the measurement of the effects of electrically active contaminants, and the instrumental analysis of ultratrace surface impurities.

While it is the purpose of this book to have experts of the various fields address these exciting developments in depth, the objective of the present chapter is to broadly review the advances achieved specifically in Si wafer cleaning technology and to outline chronologically its evolution from the 1950s to the present. Some portions of the basic material presented in the first edition of this book have been retained but are updated and expanded to reflect the new developments. Wet-chemical processes are still the most widely used method for Si wafer cleaning in the semiconductor industry today. Therefore, special emphasis is placed in treating wet processes in this chapter. Dry-cleaning processes will be briefly surveyed with to the pertinent chapters in this book.

1.2 IMPORTANCE OF CLEAN AND CONDITIONED WAFER SURFACES

The importance of clean substrate surfaces in the fabrication of semiconductor microelectronic devices has been recognized since the dawn of solid-state device technology in the 1950s. It is now well known that the device performance, reliability, and product yield of Si circuits are critically affected by the presence of chemical contaminants and particulate impurities on the wafer or device surface. Effective techniques for cleaning Si wafers before thermal treatment such as oxidation, after patterning by etching, after ion implantation, and before and after film deposition are now more important than ever before because of extreme sensitivity of the semiconductor surface and the nanometer sizes of the device features. As a consequence, the preparation of ultraclean Si wafers has become one of the key technologies in the fabrication of ULSI (ultra-large-scale integration) Si circuits.

The term “ultraclean” may be defined in terms of the concentration of both chemical contaminants and particles on the wafer surface. The International

Technology Roadmap for Semiconductors (ITRS) [1,2] specifies metrics that define the impact on yield based on contamination and defectivity for both front end of line (FEOL) processing and for complete interconnect surface preparation.

For the currrent technology nodes, the total metallic impurities should be, for example, less than 0.5×10^{10} atoms/cm^2. Particles larger than half the pitch of the smallest dimension in size should be fewer than approximately 0.1/cm^2, which translates to fewer than 30 particles per 300-mm wafer. These extremely low numbers are impressive indeed! The reason for such stringent specifications is the fact that the overall device quality is critically affected by trace impurities. Each of the hundreds of processing steps in the fabrication of advanced Si circuits can contribute to contamination. With each advancing year, the node size has been decreased to allow a more densely packed array of device features, while the device chip size has been increasing to accommodate a large number of circuits. At the same time, the permissible maximum concentrations of chemical surface impurities have been lowered. The size of particle contaminants and their density per unit area have also been specified to decrease. Therefore, the demands of surface purity are becoming continually greater and more critical, but are also more difficult to attain.

The term "surface conditioning" may need some clarification. In recent years, one has become to realize that the process of cleaning may do more than just remove impurities from the surface. It can prepare the surface chemically for the next process step. For example, immersing a Si wafer with a contaminated native (ambient) SiO_2 layer in dilute HF (dHF) solution not only removes the contaminants with the oxide film but also leaves a hydrogenated Si surface, an important prerequisite for subsequent epitaxial layer growth from the gas phase. Thus, the cleaning step has also conditioned the Si surface by a chemical reaction in preparation for the next process step.

1.2.1 Wafer Cleaning and Surface Conditioning Technology

The objective of wafer cleaning and surface conditioning is the removal of particle and chemical impurities from the semiconductor surface without damaging or deleteriously altering the substrate surface. The surface of the wafer must not be affected in such a manner that roughness, pitting, or corrosion negates the results of the cleaning process. Plasma, dry-physical, wet chemical, vapor phase, and supercritical fluid methods can be used to achieve these objectives. An extensive array of equipment is available for implementing the various processes for IC manufacturing applications.

The traditional approach of prethermal wafer cleaning and surface conditioning is based on aqueous-chemical processes that typically use hydrogen peroxide (H_2O_2) mixtures. Successful results have been achieved by this approach for the past 25 years. However, the large consumption of chemicals and rinse H_2O required by these processes and the disposal of chemical waste have led to the development of methods for reduced chemical and H_2O usage. In addition, cleaning technologies that are solvent-based, such as postetch cleaning where metal patterns are exposed and post-CMP (chemical mechanical polishing) cleaning, are evolving to use semiaqueous solvent chemistries that are more amenable to these requirements.

The development of wafer cleaning technology had a slow start in the early period of 1950−70 but then accelerated with the refinements of semiconductor device architecture and the increasing criticalness of contaminant-free surfaces. The greatly increased level of research and development for improved and new cleaning processes coupled with advances in analytical methodology and instrumentation for the detection and characterization of impurities and surface structures have been especially pronounced from 1988 to 1992. This increased level of activity is exemplified by several international conferences and symposia on wafer cleaning and related science and technology [3−8] and the appearance of several volumes on particle contamination [9−11].

Research and development from this time to the present has been experiencing an ever-increasing level of activity, as evidenced by numerous published papers and by symposia and conferences sponsored by scientific organizations. Foremost of these are the *International Symposia on Cleaning Technology in Semiconductor Device Manufacturing* whose proceedings have been published by The Electrochemical Society Inc. approximately every second year since 1990. Other important conferences have been periodically held by the Materials Research Society (MRS) whose proceedings are published as the series on *Surface Preparation Processes for Semiconductors*. Others include the *International Symposia on Ultra-Clean Processing of Silicon Surfaces*; The Fine Particle Society; *Sematech Surface Preparation and Cleaning Conference*, Semiconductor Equipment and Materials International; *MICRO*, formerly *Microcontamination*; *Semiconductor Pure Water and Chemicals Conference*; the Institute of Environmental Sciences; and the *Conferences on Chemical-Mechanical Polish Planarization for ULSI Multilevel Interconnection*. An important industry trade journal is *Solid State Technology* published by Penn Well Inc. References to specific papers will be cited in this chapter and all chapters of this book. The first few entries in references list some of these conference proceedings.

1.2.2 Wafer Cleaning and Surface Conditioning for Integrated Circuit Manufacturing

The manufacturing of an IC requires 500−800 process steps, depending on the specific type of device. Most steps are performed as unit processes with the complete wafers before dicing them into individual chips. Approximately 15%−20% of the steps are cleaning operations, which indicate the importance of cleaning and surface conditioning.

The technical jargon used in the IC surface preparation community often refers to wafers as "FEOL," "MOL," and "BEOL" to specify the stage in the processing. "FEOL" refers to wafers in the "front end of line," meaning wafers in the initial stages of processing. These wafers feature only single-crystal or polycrystalline Si (polySi) with or without SiO_2 (silicon dioxide) and Si_3N_4 (silicon nitride) layers or patterns, without exposed metal areas. Reactive chemicals with aqueous solutions can be used for cleaning and conditioning these corrosion-resistant materials. Cleaning at the early stages is typically done before gate oxide deposition and high-temperature processing, such as thermal oxidation and diffusion. The elimination of contaminants before these process steps is especially critical to prevent impurity diffusion into the substrate materials.

"MOL" refers to "middle of line" and also to MEOL (mid end of line) and is the formation of metal-based gate structures and self-aligned contacts and metal 1 layer that connects these structures. Cleaning of these structures requires penetrating high–aspect ratio structures to clean the bottom of the contacts, for example. Various dielectric materials such as high-k (high dielectric constant) and low-k dielectrics are encountered along with traditional dielectric films. Multiple metals are used in integration schemes including liners, fill layers, and capping layers. Epitaxial layers are also encountered, such as SiGe and in the future III-V semiconductors. The ability to clean without affecting the material properties is critical for the sensitive gate structure. Chemical methods involve traditional cleaning formulations, with the introduction of specialized chemicals to regulate the material loss.

"BEOL" refers to wafers in the "back end of line," later on in the processing, when interconnects are added. Cleaning of these wafers is much more restrictive because metal areas may be exposed, such as Cu (copper), Al (aluminum), or W (tungsten) metallization, possibly in conjunction with low-density or porous low-k (dielectric constant) films. Dry cleaning methods based on plasma chemistry, chemical vapor-phase reactions, and cryogenic aerosol techniques may be used to remove organic residues and particulate contaminants. Aqueous/organic solvent mixtures and other innovative approaches may also be used that will not attack exposed sensitive materials.

A general basic cleaning process flow for ULSI production can be outlined as follows with detailed chemistry formulations to be discussed in Section 1.4.2:

1. Front End of Line (FEOL)
   a. Plasma-stripping photolithographic mask and impurities after ion implantation or etching
   b. Aqueous-based residue removal and cleaning using SPM (sulfuric acid and hydrogen peroxide mixture) or O_3 (ozonated) DIW (deionized water) after plasma stripping

   - **c.** Critical surface cleaning and surface conditioning before gate oxide deposition using SC-1/SC-2 (RCA Standard Clean 1 and 2), HF (hydrofluoric acid solution) or dilute ozonated SC-1, dilute SC-2, and dHF solution before high-temperature thermal process steps
   - **d.** Post-CMP cleaning after CMP steps, using aqueous cleaning chemistry
   - **e.** Edge and back surface wafer cleaning to assure wafer cleanliness
2. Middle of Line (MOL)
   - **a.** Plasma-stripping photolithographic mask and impurities after etching
   - **b.** Aqueous-based residue removal and cleaning using dilute sulfuric acid (SPM with and without HF) after plasma stripping
   - **c.** Critical surface cleaning and surface conditioning before gate oxide deposition using SC-1/SC-2 (RCA Standard Clean 1 and 2), HF (hydrofluoric acid solution) or dilute ozonated SC-1, dilute SC-2, and dHF solution before thermal processing steps
   - **d.** Post-CMP cleaning after CMP steps, using aqueous and formulated cleaning chemistry
   - **e.** Edge and back surface wafer cleaning to assure wafer cleanliness
3. Back End of Line (BEOL)
   - **a.** Plasma-stripping photoresist mask and impurities after pattern etching
   - **b.** Poststripping cleaning using organic solvents or aqueous/organic solvent mixtures (semiaqueous solvents), in most case after plasma stripping
   - **c.** Predeposition cleaning with brush scrubber, megasonic treatment in DIW application of cryogenic aerosols, or use of supercritical CO_2 (carbon dioxide)
   - **d.** Post-CMP cleaning after CMP steps, using aqueous and formulated cleaning chemistry
   - **e.** Edge and back surface wafer cleaning to assure wafer cleanliness

The distribution of cleaning and surface conditioning steps in the process flow for manufacturing a typical 14-nm logic IC in 2017 is presented in Table 1.2-1. It can be seen that approximately 70% of all cleaning steps are based on aqueous chemistries and 30% are based on plasma processing. There are a small number of cleaning steps that are vapor-based or use other dry cleaning methods, not involving plasma or aqueous-based processing.

A specialized cleaning operation is required for Si device wafers after CMP. Depending on the metallization used for the device, the exposed surface may consist of Al, Cu, or W in addition to Si, SiO_2, silicate glasses and doped glasses, SiN_x, Si_3N_4, diffusion barrier layers such as TaN and TiN, etch stop films, and possibly other materials. The primary purpose is the removal of particles from the polishing slurry, which may contain Al_2O_3 (aluminum dioxide), SiO_2, CeO_2 (ceric oxide), and other types of abrasives and chemicals. The approach taken is usually the application of brush scrubbing and/or megasonic cleaning in combination with noncorrosive aqueous-based chemical solutions.

TABLE 1.2-1 Distribution of Cleaning Steps in the Manufacturing Process Flow of a Typical 14-nm Logic IC in 2017

Cleaning Step	Total Number of Steps
BEOL aqueous post-strip	22
BEOL plasma clean	22
BEOL premetallization clean	21
FEOL aqueous post-strip	15
FEOL plasma strip	15
FEOL aqueous critical clean	10
FEOL and BEOL defectivity improvement (particle removal)	10
FEOL and BEOL post-CMP clean	14
Total	**119**

1.3 OVERVIEW OF WAFER CONTAMINATION ASPECTS

The contamination of Si wafer surfaces is intimately associated with wafer cleaning and surface conditioning because this is the very reason for cleaning. Chapter 2 discusses this subject in depth, covering both chemical and particle contamination and the resulting defectivity. Key aspects of contamination are briefly outlined in this section, which may serve as an introduction to Chapter 2.

1.3.1 Types and Origins of Contaminants and Defectivity

Contaminants on wafer surfaces exist as adsorbed ions and elements, thin films, discrete particles, particulates (clusters of particles), and adsorbed gases. Surface contaminant films and particles can be classified as molecular compounds, ionic materials, and atomic species. Molecular compounds are mostly particles or films of condensed organic vapors from lubricants, greases, photoresists, solvent residues, organic compounds from DIW, fingerprints or plastic storage containers, and metal oxides or hydroxides. Ionic materials comprise cations and anions, mostly from inorganic compounds that may be physically adsorbed or chemically bonded (chemisorbed), such as ions of Na (sodium), F (fluorine), and Cl (chlorine). Atomic or elemental species comprise metals, such as Cu and heavy metals that may be electrochemically plated out on the semiconductor surface from HF-containing solutions, or they may consist of Si particles, dust, fibers, or metal debris from equipment.

Particles can originate from airborne dust from equipment, processing chemicals, factory operators, gas piping, wafer handling, and film deposition systems. Mechanical (moving) equipment and containers for liquids are especially prolific sources, whereas solid materials, liquids, gases, chemicals, and ambient air tend to cause less particle contamination, but all contribute significantly to the generation of chemical contaminants. Static charge build up on wafers and carriers is a powerful force for particle deposition that is often overlooked and should be neutralized by proper grounding.

A selection of literature references relating to the sources of contaminants on Si surfaces is as follows: on airborne molecular contamination [12]; organic contaminants from clean room air [13]; metals from HF etchants [14]; metal ions from aqueous solutions [15]; cations from ammonia-containing solutions [16]; metal impurities from RCA SC-1 solution [17]; metal impurities and from alkaline solutions [18]; particles from dHF cleans [19]; and particles from electrostatic attraction [20].

1.3.2 Types of Semiconductor Wafers

Front end of the line Si wafers can be in the form of mechanically lapped and chemo-mechanically polished slices cut by saw from single-crystal ingots. They may be coated with an epitaxial layer of Si with different dopant types and concentrations, and/or they may be coated with a film of uniform or patterned Si dioxide. Up to this point in the processing, wafer cleaning operations can utilize highly reactive chemicals that do not attack these corrosion-resistant materials. The situation changes once layers or patterns of deposited metals are present on the BEOL wafers. As already noted, the cleaning chemistry must then be confined to mild and noncorrosive treatments, such as rinses with dilute acids. Alternatively, dry-cleaning methods such as cryogenic aerosols or supercritical fluids can be applied.

In this connection, we should mention that a small fraction of wafers in the fabrication of solid-state microelectronic devices utilize compound semiconductors such as gallium arsenide (GaAs), gallium phosphide (GaP), and numerous complex alloys. The importance of these materials has grown steadily for unique applications in optoelectronics. The basic differences in chemical properties between these semiconductors and those of Si require different and specialized cleaning treatments that are outside the scope of this book.

1.3.3 Effects of Contaminants and Defectivity on Silicon Device

The various effects of contaminants on semiconductor Si, layers of dielectrics and insulators, and finished semiconductor devices are complex and depend on the nature, location, and quantity of the specific type of impurity. Purification of wafer surfaces to remove contaminants is essential because over 50% of the yield losses in the IC manufacturing are estimated to be caused by microcontamination. A selection of representative early papers published from 1987 to 1992 on effects of contaminants are included in Refs. [21−40].

More recent representative papers on the effects of contaminants on Si and dielectrics include carbon contamination on device performance [41]; organics on electrical degradation [42]; iron (Fe) on surface roughening [43,44]; electrical effects of cobalt (Co) and Cu [45]; Cu on thin oxide breakdown [46], on gate dielectrics [47,48], in bulk Si [49]; and ionic contaminants on thermal oxidation of Si [50]. Two excellent surveys on the effects of numerous metal contaminants were presented in Refs. [51,52].

Molecular contaminant films on wafer surfaces can act as a mask and thereby prevent effective cleaning or rinsing, impair good adhesion of deposited films, and on heating form deleterious decomposition products. For example, organic residues, if exposed to high temperatures in a nonoxidizing atmosphere, can carbonize and form silicon carbide (SiC) that can nucleate polySi regions in an epitaxial layer.

Ionic contaminants cause a host of problems in Si devices. During high-temperature processing or on application of an electrical field, they may diffuse into the bulk of the semiconductor structure or spread on the surface, leading to electrical defects, device degradation, and yield losses. For example, highly mobile alkali ions in Si and amorphous SiO_2 films on Si may cause drift currents and unstable surface potential, cause shifts in threshold and flat-band voltages, lead to surface current leakage, and lower the oxide breakdown field of thermally grown or deposited films of SiO_2. In the growth of epitaxial Si layers, sufficiently high concentrations of ions can give rise to twinning dislocations, stacking faults, and other crystal defects.

Certain metallic contaminants are especially detrimental to the performance of Si devices, as indicated in Table 1.3-2. Because Si is above H in the electromotive series of the elements, heavy metals tend to deposit from aqueous solutions on its surface by galvanic action, actually plating out with high efficiency, especially from HF-containing etchants. If not removed, these impurities may diffuse into the Si substrate during subsequent heat treatments and introduce energy levels into the forbidden band to act as traps or

TABLE 1.3-2 Critical Impurity Elements for Si Devices. The Following Impurity Elements From Chemicals and Processes in the Manufacturing of Si ICs Have Deleterious Effects on Device Performance

Heavy metals (most critical)	Cu, Fe, Ni, Cr, Co, Mo
Alkali metals (critical)	Na, K, Li, Ca
Other elements (least critical)	Al, Mg, C, S, Cl, F

generation/recombination centers, cause uncontrolled drifts in the semiconductor surface potential, affect the surface minority-carrier lifetime and the surface recombination velocity, lead to inversion or accumulation layers, cause excessive leakage currents, and give rise to various other device degradation and reliability problems. Metal contaminants in or on Si wafers can lead to structural defects in vapor growth of epitaxial layers and degrade the breakdown voltage of gate oxides.

Particles can cause blocking or masking of wafer processing operations, such as photolithography, etching, deposition, and rinsing. They may obstinately adhere to surfaces by electrostatic adsorption and may become embedded during film formation. Deposition and removal of particles becomes exacerbated as the size decreases because of the extremely strong adhesion forces. Furthermore, particles constitute a potential source of chemical contamination, depending on their composition. Particles that are present during film growth or deposition can lead to pinholes, microvoids, microcracks, and other structural defects depending on their chemical composition. In advanced stages of device fabrication, particles can cause shorts between conductor lines if they are sufficiently large, conductive, and located between conductor lines. They are considered potential device killers if their size is larger than one-tenth the size of the smallest feature of the affected IC.

1.3.4 Prevention of Contamination and Defectivity

Surface microcontamination and its control [9], prevention detection, and measurement are important topics that are addressed in Chapter 2 and are only briefly discussed in this section. The key notion of this subject is *prevention*. If we can prevent contamination during the entire device manufacturing process by creating and maintaining superclean conditions in equipment, materials, and environment, there would be little need for wafer cleaning. Furthermore, it is generally easier (or less difficult!) to prevent contamination rather than to remove it once it has occurred. Therefore, avoiding contamination must be the first priority, and strict contamination control should be exercised to the fullest extent throughout device manufacturing [53]. Contamination control requirements were described [26,54] and have been reassessed more recently [54−61].

Processing equipment has been the major source of particle contamination [62] and must be controlled effectively by eliminating the dust through scheduled maintenance and by electrostatic charge removal [20,62]. Particle generation can be further minimized by reducing friction of moving equipment parts, avoiding turbulent gas flows, reducing operator handling through automation, and by exercising periodic cleanup [63].

Next to equipment, semiconductor wafer processing in the fab must be controlled. Carefully optimized processing conditions for film deposition, plasma etching, ion implant, thermal treatment, and other critical processing steps are effective measures for preventing particle deposition. For example, in chemical vapor deposition, particle-generating homogeneous gas-phase nucleation must be minimized by optimizing the reaction parameters. Recirculation of the partially depleted reaction gases must be avoided by improved equipment design. Sudden burst in the introduction of gases into a system should be prevented by using "soft starts" to increase the gas flow gradually, and so forth [64].

Prevention of contamination by particles and chemical impurities should be exercised by implementing ultrafiltration of gases, DIW, and processing liquids. Point-of-use (POU) final purification is especially effective to accomplish this requirement. Furthermore, continuous efforts have been made by the producers of chemicals to provide the semiconductor electronics industry with ultrapure chemicals [65−68]. "Ultrapure" may be defined as total impurity concentrations in the ppt (parts per trillion) range. An interesting alternative approach to create ultrapure chemicals is the POU generation using gaseous chemicals (O_3, HCl, and NH_3) with DIW [69−72]. Finally, DIW, which is so extensively used throughout wafer cleaning and rinsing operations, must be effectively purified to prevent surface contamination [66,73,74].

Refined methods for the analysis of trace impurities on Si surfaces are available [75−84] to monitor the results of cleaning processes, as discussed in Chapter 10.

1.4 OVERVIEW OF WAFER CLEANING AND SURFACE CONDITIONING TECHNOLOGY

The topics discussed in this section include liquid processes, gas-phase methods, and wafer rinsing and drying techniques. References to the literature here and elsewhere in this chapter are intended to be representative rather than comprehensive.

1.4.1 Liquid Processes and Wafer Drying Techniques

Liquid processes for wafer cleaning and surface conditioning are based on the use of aqueous chemicals, organic solvents, or mixtures of the two. If aqueous chemicals are used, the process is properly called "wet-chemical." Most of these processes are for FEOL wafers, unless specifically noted for applications to BEOL wafers. Wet-chemical processes are still the primary workhorse in Si wafer cleaning and are therefore given special emphasis in this chapter.

The mechanism of liquid cleaning can be purely physical dissolution and/or chemical reaction dissolution. Chemical etching occurs when materials are removed by a chemical transformation to soluble species. Traditionally, chemical etching is expected to remove substantial quantities of a material, such as a deposit film on a substrate. However, certain chemical cleaning processes may result in the removal of only a few atom layers of material (as in

SC-1 cleaning) and by above definition should also be considered chemical etching; perhaps the term *microetching* would be a more appropriate description. Conventional liquid-chemical etching processes for the removal of bulk quantities of electronic materials is beyond the scope of this book. Several comprehensive reviews have been published that cover this topic thoroughly [85−88].

Liquid methods of wafer cleaning are based on the application of mineral acids, aqueous solutions including H_2O_2-containing mixtures, organic solvents, and aqueous/solvent combinations. Different process sequences are used for specific applications. A variety of technical equipments are available commercially for efficiently implementing cleaning processes for high-volume fabrication of IC devices. Rinsing, drying, and storing of cleaned wafers are intimately related to cleaning operations and are addressed in the section that follows.

Organic solvents are rarely used for cleaning premetallized FEOL Si wafers because much more effective cleaning agents can be used. BEOL device wafers, however, are sometimes treated with organic solvents to attain some degree of cleaning, because wet-chemical cleans cannot usually be used unless specially formulated. Chlorofluorocarbon compounds, acetone, methanol, ethanol, and isopropyl alcohol (IPA) are solvents that are frequently used to remove organic impurities. IPA is generally the purest available organic solvent and is used extensively for vapor drying of H_2O-rinsed wafers.

1.4.2 Wet-Chemical Cleaning Processes

1.4.2.1 Hydrofluoric Acid Solutions

Mixtures of concentrated hydrofluoric acid (49 wt% HF) and DIW have been widely used for etching films of silicon dioxide (SiO_2), silicate glasses (e.g., phosphosilicates, borophosphosilicates), and silicon nitride (Si_3N_4, SiN_x) that were grown or vapor deposited on Si substrate wafers. The chemical dissolution reactions have been identified and described in the literature [85,86].

The thin layer of native oxide on Si, typically 1.0−1.5 nm thick, is removed by a brief immersion of the wafers in diluted (typically 1:50 or 1:100) ultrapure and POU filtered HF solution at room temperature. The change of the wetting characteristics of the initially hydrophilic to a hydrophobic surface, which strongly repels aqueous solutions, can visually indicate when the oxide dissolution is complete. The effect is caused by the H-passivated Si surface that results from exposure to HF. The resulting oxide-free H-terminated Si surface strongly attracts oppositely charged particles and is very sensitive to organic contaminants from DIW and ambient air. Therefore, etching with HF solutions that leave the Si surface bare must be carried out with very dilute ultrapure (Fe-free) and ultrafiltered HF solution in a very low-particle atmosphere. In addition to etching, HF solutions can desorb metallic impurities from the Si surface. If desired, a pure SiO_2 film can be regrown by exposure of the HF treated Si to an oxidizing solution.

Mixtures of HF and ammonium fluoride (NH_4F) solutions are known as buffered oxide etch (BOE or BHF) and are used for pattern delineation etching of oxide and glass films. Buffering results in a more stable etch rate and prevents loss of photoresist polymer films that cannot withstand strongly acidic unbuffered HF solutions [89]. Surfactants may be added to the etchant to improve the wetting characteristics or to prevent microroughness of the Si surface [90]. Whereas the free acid is the major etching species in aqueous HF, the ionized F species HF_2^- is the major etchant component in buffered HF solutions. Addition of NH_4F increases the pH to 3−4, maintains the concentration of F^-, stabilizes the etching rate, and produces the highly reactive HF_2^-. A commonly used BHF volume ratio of 7:1 NH_4F (40 wt%)−HF (49 wt%) has a pH of about 4.5 and appears to contain only HF_2^- and F^-, with very little free HF acid. The SiO_2 etching rate of HF_2^- is four to five times as fast as that for HF species in aqueous HF acid [86]. Additional details of the complex chemistry in these reactions (and or liquid-chemical etching in general) can be found in Refs. [85−89,91−96]. The subtle differences in the Si surface morphology resulting from these two types of fluoride etchants are discussed in detail in Chapter 9.

1.4.2.2 Sulfuric Acid and Hydrogen Peroxide Mixtures

Removal of gross organic materials from Si wafers, such as hardened photoresist polymer patterns after ion implantation and other visible contaminants of organic nature, can be accomplished by using mixtures of 98 wt% H_2SO_4 (sulfuric acid) and 30 wt% H_2O_2. Volume ratios of 2:1−4:1 are used at a temperature of 100−130°C for 10−15 min. Organics are destroyed and eliminated by wet-chemical oxidation, but inorganic contaminants, such as metals, are not desorbed. What is worse, the Si surface after this cleaning step is strongly contaminated with S (sulfur) residues from the H_2SO_4 [94−97]. These SPM, which are also known as "piranha etch" (because of their voracious ability to eradicate organics) or, incorrectly, "Caros acid," are extremely dangerous to handle in the fab; goggles, face shields, and plastic gloves are needed to protect the operators. This procedure is usually the first cleaning process to prepare grossly contaminated FEOL Si wafers for subsequent treatments. Vigorous rinsing with DIW is required to completely remove the viscous liquid. Finally, it is advantageous after the DIW rinsing to strip the impurity-containing formed oxide film on Si or on the thermal SiO_2 layer by dipping the wafers for 15 s in HF−H_2O (1:50), followed by a DIW rinse. A modification can be made by adding minute amounts of HF to the SPM, which results in better removal of sulfur compounds, shorter rinsing times, and improved particle removal [94].

Some alternative oxidants are sometimes used instead of H_2O_2 in an effort to improve the stability of the mixtures. These alternative additives include ammonium persulfate $(NH_4)_2SO_4$, known as SA-80 when combined with

H_2SO_4 [98]; peroxydisulfuric acid ($H_2S_2O_8$), called "Caros acid" when combined with H_2SO_4 [99]; and ozonated H_2O, DIW/O_3, called sulfuric acid—ozone mixture (SOM) when combined with H_2SO_4 [100].

1.4.2.3 Original RCA Cleaning Process

The first successful process for wet-cleaning FEOL Si wafers was systematically developed at RCA, used for several years in the fabs, and finally published in 1970 [101]. The process consists of two consecutively applied hot solutions known as "RCA Standard Clean," SC-1 and SC-2, featuring pure and volatile reagents. These solutions have been widely used in their original or modified form for over 40 years in the fabrication of Si semiconductor devices. The SC-1 solution for the first processing step consists of a mixture of NH_4OH (ammonium hydroxide), H_2O_2, and H_2O, also known as "ammonia—peroxide mixture. The SC-2 solution for the second processing step consists of a mixture of HCl (hydrochloric acid), H_2O_2, and H_2O, also known as "hydrochloric—peroxide mixture." The process as *originally* published [101] is concisely described below. Changed or modified versions of the RCA clean as they are used now will be discussed in the next section.

The *originally* specified composition for the SC-1 solution ranges from 5:1:1 to 7:2:1 parts by volume of H_2O:H_2O_2:NH_4OH. Filtered DIW or quartz-distilled H_2O is used. The H_2O_2 is electronic grade 30 wt% H_2O_2, unstabilized (to exclude contaminating stabilizers). The NH_4OH is 27% (wt/wt% based on NH_3). The originally specified composition for the SC-2 solution ranges from 6:1:1 to 8:2:1 parts by volume of H_2O:H_2O_2:HCl; the H_2O and H_2O_2 are as noted above. The HCl concentration is 37 wt%.

The exact compositions for both solutions are not critical for proper performance; the recommended proportions are reliable and simple to prepare and use. Cleaning in either mixture is carried out at 75—85°C for 10—20 min followed by a quench and overflow rinse in running DIW. Diluting the hot bath solutions with cold H_2O is done to displace the surface level of the liquid and to reduce the bath temperature to prevent any drying of the wafers on withdrawal from the bath. The wafers are rinsed with running DIW or quartz-distilled H_2O and then spun dry. They are immediately transferred to an enclosure flushed with prefiltered inert gas for storage, if they cannot be processed immediately.

The SC-1 solution was designed to remove from Si, oxide and quartz surfaces organic contaminants that are attacked by both the solvating action of the NH_4OH and the powerful oxidizing action of the alkaline H_2O_2. The NH_4OH also serves to remove by complexing some periodic group IB and IIB metals such as Cu, Au, Ag, Zn, and Cd, and some elements from other groups such as Ni, Co, and Cr. In fact, Cu, Ni, Co, and Zn are known to form amine complexes. It was not originally realized that, before the ability to perform AFM (atomic force microscope) analysis, SC-1 dissolves the thin native oxide layer on Si at a very low rate and forms a new oxide on the Si surface by oxidation at approximately the same rate. This oxide regeneration is now believed to be an important factor in the removal of particles and chemical impurities.

The SC-2 solution was designed to dissolve and remove from the Si surface alkali residues and any residual trace metals (such as Au and Ag), and metal hydroxides, including $Al(OH)_3$, $Fe(OH)_3$, $Mg(OH)_2$, and $Zn(OH)_2$. Displacement replating from solution is prevented by formation of soluble metal complexes with the dissolved ions. The solution does not etch Si or SiO_2 and does not have the beneficial surfactant activity of SC-1 for removing particles. SC-2 has better thermal stability than SC-1 so that the treatment temperature need not be as closely controlled.

The original paper [101] demonstrated the effectiveness of the process for the removal of organic surface contaminants by sensitive H_2O spray tests, the elimination of electrically active surface impurities by electrical measurements of MOS devices, and by radiochemical tracer studies of adsorption and removal of various metallic nuclides. The relative oxidizing powers of the cleaning solutions were compared on an electrode potential versus pH diagram, demonstrating the remarkable activity of the solutions to solubilize most metal adsorbates by oxidation and complexing. The thermal instability of H_2O_2 in SC-1 (and to a lesser extent in SC-2) necessitates the use of freshly prepared mixtures. The effects of H_2O_2 depletion and of fluoride ion addition were also investigated in terms of Si and SiO_2 etch rates.

Several improvements to the original RCA cleaning procedure were reported by Kern in a number of papers [102−107]. The most influential of these changes was the introduction of the RCA megasonic cleaning system for cleaning and rinsing of wafers [104]. Megasonic treatment in an SC-1 bath is especially advantageous for physically dislodging particles from the wafer surface because of the high level of kinetic energy. It allows a substantial reduction in solution temperature and offers a much more efficient mode of rinsing than immersion tank processing.

Other important improvements include the simplification of the composition ratios for both SC-1 and SC-2 to 5:1:1 and a reduction of treatment temperature and time to 70−75°C for just 5−10 min. The use of fused silica vessels instead of Pyrex glassware was introduced to eliminate contamination from leached glass components. The separation of SC-1 from SC-2 processing stations in two dedicated exhaust hoods was recommended to avoid contamination from airborne colloidal NH_4Cl generated from NH_3 and HCl vapors. No gross etching of Si or oxide was shown to occur with SC-1 even if the H_2O_2 concentration is reduced by a factor of 10. Finally, an optional process step was introduced by stripping the hydrous oxide film formed after SC-1 with high-purity particle-free, 1:50 dHF for 10 s so as to reexpose the Si surface for the subsequent SC-2 step.

1.4.2.4 Modifications of the RCA Cleaning Process

The development of several improvements in RCA cleaning has been reported in the early literature and relates mainly to four areas of processing:

1. Two alternative implementation techniques are centrifugal spray processing where the cleaning solutions are sprayed on spinning wafers in a centrifuge [108]; and stationary closed-system processing where the wafers remain enclosed in a module during the entire cleaning, rinsing, and drying procedure [109].
2. Reduction of the NH_4OH concentration in SC-1 by a factor of 2−10 to prevent microroughening of the Si surface and to enhance particle removal, thereby improving the quality of gate oxides [35−40].
3. Dilution of SC-1 and SC-2 with DIW to various concentrations while still achieving good cleaning effectiveness [110].
4. Replacement of SC-2 with very dilute, room-temperature HCl because Au and Ag contaminants are now no longer present in high-purity process chemicals. Previously, any deposits of Au or Ag on Si required SC-2 for oxidative desorption. Other residual metals and their hydroxides are readily soluble in dilute HCl [111] (preferably containing some O_2).

1.4.2.5 Alternative Cleaning Solutions

Several additional solutions for wet-chemical cleaning and surface conditioning are available for use in conjunction with or as a replacement for RCA Standard Cleaning and will be briefly surveyed:

HF-last processing is used for creating an H-passivated, hydrophobic Si surface by briefly immersing cleaned wafers as a last step of the RCA sequence in diluted ultrahigh purity HF, followed by final rinsing and drying [112−115]. Alternative to wet processing, the wafers can be exposed to HF-IPA vapor [113].

Dilute chemistry refers to the dilution of cleaning solutions to various concentrations. Optimal dilution levels have been established for specific process steps to achieve maximum cleaning efficiency at minimal consumption of chemicals [110,116−119]. The degree of dilution can be varied greatly and is sometimes carried to extremes. For example, a 1000:1:1 ratio for "ultra-diluted" SC-1 in conjunction with a nonionic surfactant and megasonic treatment at 65°C has been reported for successful wafer cleaning applications [119]. However, certain compromises in optimization should be carefully considered to avoid negative effects [120].

Ozonated solutions have become an important part in the cleaning chemistry. The use of ozonated DIW (O_3/DIW) was shown in one of the earliest reports to offer substantial advantages over cleaning mixtures prepared with H_2O_2 [100]. The authors generated O_3/DIW in the range of 10−15 ppm, added it to H_2SO_4, and applied the mixture (SOM) by centrifugal spraying. After the second step with dHF, spray rinsing was done with O_3/DIW. Since then, many uses for O_3 in wet cleaning of Si wafers have been introduced, as noted in selected Refs. [72,100,121–132].

Ozone is one of the strongest oxidants available; it generates an active radical when dissolved in ultrapure H_2O (UPW). For comparison, the oxidation–reduction potential of O_3 is 2.07 V, whereas for H_2O_2 it is only 1.78 V. Major advantages of O_3 include reduction of chemical usage, lowered processing costs, improved cleaning performance, and a high quality of oxide if used for forming SiO_2 films [122]. Removal of organic contaminants can be accomplished at room temperature with O_3/DIW [122–125]. SOM has been used to replace the traditional SPM process for stripping photoresist polymer films [126]. An optimized process using O_3/DIW with ammonium bicarbonate (NH_4HCO_3) can remove photoresist layers on BEOL wafers with Al metallization [127].

Ozonization has also been done by injecting O_3 into various liquid chemicals other than DIW and H_2SO_4 [72,123], including dHF for removing both resist and postetch polymer residues [128]. Postmetal etching resist or residue has been shown to be removable from BEOL wafers with O_3/DIW at 50°C [129]. The chemistry involved in the reactions of O_3 with NH_4OH and Si has been investigated [130], and the reactions involved in the removal of organics in O_3/DIW [131]. As mentioned already, an important additional use of ozonated UPW is for forming thin oxide films on Si after exposure to HF [122,132].

Micro-etching solutions based on dHF and an oxidant, such as H_2O_2 or HNO_3, can have better efficiency than conventional cleans for removing Cu and other metallic surface contaminants, if some loss of Si surface layer is acceptable [14,115,121,133–138]. For example, a typical mixture of 1% H_2O_2–0.5% HF has been used [133], and one of 0.025%–0.1% HF–HNO_3 [134].

Miscellaneous cleaning solutions have been reported that can offer advantages over conventional cleaning. The addition of dilute HCl to dHF enhances the removal of Cu (and possibly other trace metal deposits) [121,136,137,139,140]. A different type of cleaning composition uses choline (trimethyl-2-hydroxyethyl ammonium hydroxide) instead of NH_4OH in SC-1 [141–144], with possibly improved performance [143]. A similar commercial composition utilizes tetramethyl ammonium hydroxide ($(CH_3)_4NOH \cdot 5H_2O$) as an alternate organic base, with H_2O_2 in a spray application [145].

An aqueous solution of hydroxylamine (NH_2OH) at 60–70°C has been reported for removing postetch polymer residues from BEOL wafers without structural damage [146,147]; alternatively, semiaqueous solvents can be used advantageously with the same reagent [147].

The application of surfactant additives to aqueous solutions has become widely accepted. Surfactants increase the wettability of Si wafers, aid in the removal of fine particles, can prevent microroughness generation of Si during etching, and improve the overall cleaning efficiency [14,55,90,119,136,137,148–153].

They have been added to both alkaline and acidic media including SC-1, SC-2, dHF, dHF-H_2O_2, and dHF-O_3. Anionic, cationic, and nonionic types have been applied to modify the polarity of the zeta potentials in minimizing particle deposition.

Finally, chelating or complexing agents have been shown to enhance the cleaning activity of aqueous solutions and prevent adsorption and readsorption of dissolved metal contaminants [17,119,149,154−159]. These agents are much more chemically reactive than the surfactants noted above. The oldest and best-known agent is EDTA (ethylendiamine tetraacetic acid) for binding Ca ions, but numerous other chemicals have been studied as additives to aqueous cleaning solutions with beneficial results.

1.4.3 Alternative Wet-Chemical Cleaning and Surface Conditioning Systems

Very extensive and excellent research has been done in recent years by several groups of investigators to develop wet-chemical Si wafer cleaning and surface conditioning processes that would have advantages over the traditional RCA-type cleaning. These new processes typically combine several of the discrete cleaning steps described in the previous sections to create a consolidated and optimized procedure. All of these alternative procedures are based on wet-chemical cleaning, which is favored because many inherent properties of aqueous solutions facilitate a broad-spectrum removal of metal deposits, particles, and other contaminants, which would not be possible with dry cleaning methods. The reasons for changing the wet chemistry of cleaning have been the need to generate improved high-performance gate dielectrics, which require ultrahigh purity Si surfaces, and cost reduction of the entire cleaning procedure. These alternate cleaning systems have become known as Ohmi Clean, IMEC Clean, Diluted Dynamic Clean, and single-wafer/short-cycle clean. The salient features of each are described below.

1.4.3.1 Ohmi Clean

Ohmi Clean refers to the systems developed over many years by the research group at Tohoku University in Sendai, Japan, under the leadership of Professor T. Ohmi. The features of these systems include the application of dilute chemistry with HF, H_2O_2, ozonated and hydrogenated ultrahigh purity H_2O (UPW), and fast processing at room temperature with megasonics with or without surfactants in an optimized processing sequence [114,138,160−162]. A typical procedure for FEOL wafer cleaning is shown in Fig. 1.4-1 [114,160].

A simplified four-step system was developed more recently after a series of intermediate ones [161]. It does not use surfactants and employs O_3/UPW and H_2/UPW with megasonics, dHF-H_2O_2, and H_2/UPW rinsing with megasonics followed by drying. Details are shown in Fig. 1.4-2 [161,162].

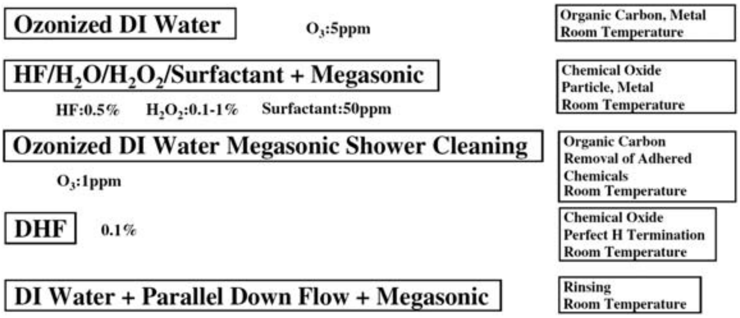

FIGURE 1.4-1 Wet cleaning process developed by professor Ohmi [160]. *Reprinted with permission of SCP Semiconductor Technologies.*

A two-step room temperature wet-cleaning procedure for BEOL wafers is based on dHF with an anionic surfactant. The etch rate of some metallic films is controlled by adding 0–0.5 vol% H_2O_2. Rinsing is done with O_3/UPW to remove surfactant residues. An outline of this interesting process is presented in Fig. 1.4-3 [162].

1.4.3.2 IMEC Clean

A very active group of researchers at IMEC in Leuven, Belgium, have been performing outstanding pioneering work in developing cleaning technologies for metal and particle removal preliminary to preparing critical oxide layers; they have published their results in numerous presentations and papers [52,113,163–165].

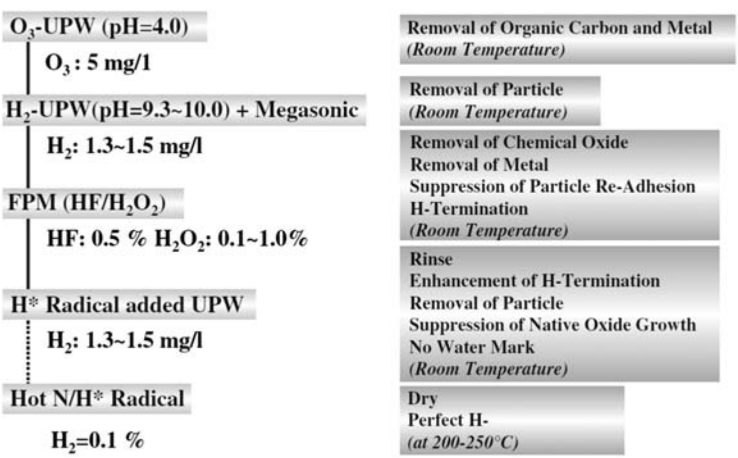

FIGURE 1.4-2 Four-step room temperature wet-cleaning for FEOL [161,162]. *Reprinted with permission of SCP Semiconductor Technologies.*

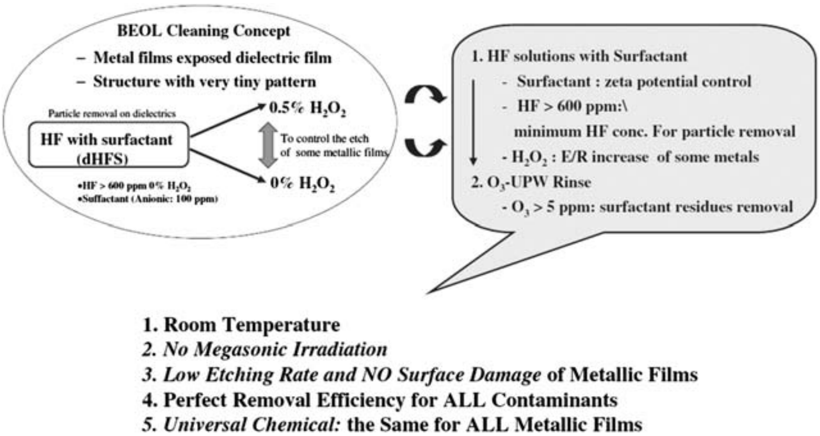

FIGURE 1.4-3 Two-step room temperature wet cleaning for BEOL [162]. *Reprinted with permission of SCP Semiconductor Technologies.*

A roadmap for simplifying the cleaning process from RCA clean to IMEC Clean and eventually to single-wafer wet clean is outlined in Fig. 1.4-4 as a guide [165]. The IMEC Clean concept is based on a two-step cleaning approach and is illustrated in Fig. 1.4-5.

The bold-printed insets are the preferred choices over alternative ones indicated. The purposes for each process are shown below each step. If the

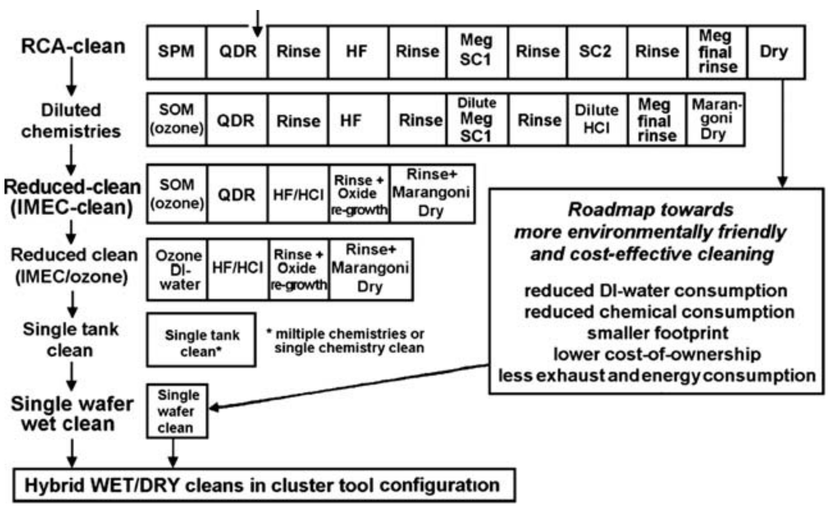

FIGURE 1.4-4 Suggested roadmap toward cleans with lower chemical and DIW consumption [165]. *Reprinted with permission of SCP Semiconductor Technologies.*

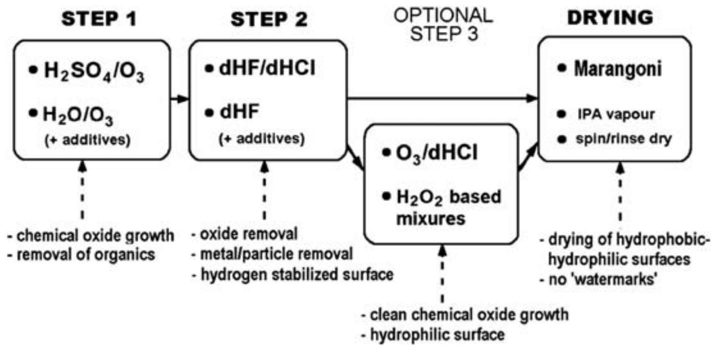

FIGURE 1.4-5 Schematic illustration of the IMEC Clean concept [165]. *Reprinted with permission of SCP Semiconductor Technologies.*

IMEC Clean recipe is implemented in an automated wet bench, the total processing time for a batch of wafers is typically 32 min. Excellent cleaning performance has been demonstrated with this cost-effective replacement for the traditional RCA clean [165].

1.4.3.3 Diluted Dynamic Clean

This system, named dilute dynamic clean, "DDC," has been developed mainly by researchers at Gressi-Leti in France [166–168]. It uses only diluted chemicals at room temperature and can be optimized for removal of SiO_2 films, particles, or metal deposits [166], and for pregate cleaning applications [168]. A conventional quartz overflow rinsing tank is used into which small amounts of O_3 and/or gaseous HCl are injected into the continuous flow of DIW. Both the cleaning and rinsing are performed alternatively in the same tank. A second tank is dedicated to HF-based chemistry (1 wt% HF) featuring a recirculated/filtered bath solution with an O_2 desorption device and a chemical purifier to remove noble metals. The quantity of chemicals consumed in DDC is reduced more than 10-fold over the conventional SPM/HF/RCA clean process and the cycle time is decreased twofold, with results as good as with RCA clean [168].

1.4.3.4 Single-Wafer/Short-Cycle Clean

Single-wafer processing has become more important in recent years because it allows superior processing control, prevents cross-contamination, improves technical performance, and facilitates better integration with other processing steps than conventional batch processing. Other important advantages are the greatly decreased cycle time and reduction in the volume of cleaning chemicals and DIW while still resulting in good effectiveness. One system employs

O_3/DIW and dHF in an alternating repetitive sequence. The solutions are dispensed over the horizontally spinning wafer for a few seconds each. The sequence is repeated as many times as necessary until the wafer surface has attained the desired degree of cleanliness. Spin drying is done in a N_2 atmosphere. Excellent results in cleaning efficiency have been reported [169−171].

Another FEOL wet-cleaning procedure consists of a two-step spin/spray with diluted SC-1 for 10 s followed by a rinse and a 10-s dHF−HCl spray [172].

A short-cycle, single-wafer cleaning system ("AM Clean") features a horizontal spin unit with a dispense/spray setup for 300-mm wafers. It uses a treatment with 30 s dHF, a 20-s UPW rinse, a 30-sec modified SC-1 spray/rinse, and a 20-s final rinse followed by 20 s of centrifugal drying. The modified SC-1 contains a surfactant and a chelating agent. The entire cycle time requires just 2 min instead of 64 min for conventional immersion cleaning, as shown in Fig. 1.4-6 [173,174].

A full-coverage megasonic system associated with the single-wafer horizontal spin unit can provide uniformly focused acoustic energy over a wafer for improved efficiency [174]. Excellent results have been published with this novel wafer cleaning system.

Finally, a single-wafer tool for immersion wet processing delicate sub−100-nm device structures has been described which uses dilute SC-1 solution and a final dHCl rinse. A multiple megasonic transducer array was built in the vertical process chamber to direct acoustic energy to the wafer surface. Process and rinse times of 30 s are typical. IPA vapor drying has been successfully incorporated with reportedly good results [175].

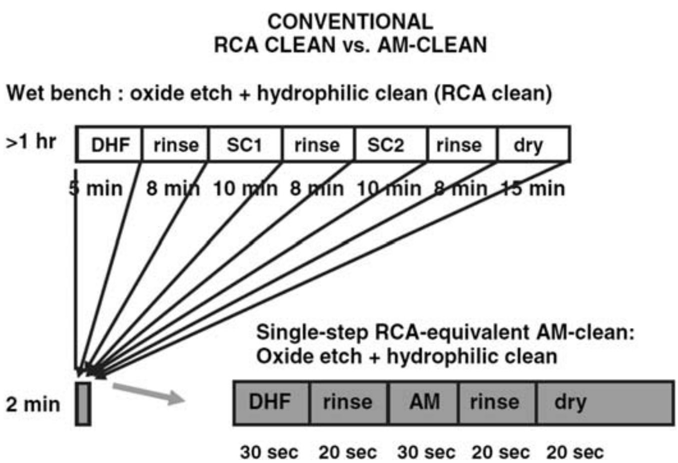

FIGURE 1.4-6 Performance of a novel single-wafer/short-cycle wafer cleaning system [174]. *Used with permission from Semiconductor International.*

1.4.4 Equipment for Implementing Wet-Chemical Cleaning

A large choice of commercial equipment for implementing wet-cleaning processes is available. The selection of a particular type of equipment depends on many factors, such as batch- or single-wafer processing, wafer size and device type, throughput requirements, and FEOL or BEOL processing. The various types of equipments are briefly described below and will be discussed in greater detail in Chapter 4.

The original RCA cleaning process utilized a *simple immersion tank* of fused quartz, or plastic polymer material for HF solutions. Wafers were placed in a slotted plastic carrier, which was immersed in the bath solutions. *Commercial tanks* feature an overflow and recirculation setup with temperature control. *Immersion tank wet benches* for large-scale batch processing are now produced by several equipment manufacturers. These units are quite elaborate, featuring a series of tanks for different bath solutions, all enclosed with HEPA (high-efficiency particulate air)-filtered airflow, and a fully automated robotic processing capability. They may also incorporate megasonic and IPA vapor drying stations.

Centrifugal spray machines for batch- or single-wafer dynamic wet-cleaning use freshly mixed and separately introduced cleaning solutions. They are more conservative on chemical and H_2O use and allow cleaning, rinsing, and spin-drying or IPA vapor drying in the same unit, with greatly reduced time cycles.

Static closed systems designed for batch- or single-wafer processing retain the wafers in an enclosed module for the entire cleaning, rinsing, and drying sequence. The hot or cold solutions and IPA are introduced by programmed direct displacement techniques.

Megasonic systems produce ultrahigh frequency energy at 0.85−0.90 MHz generated by piezoelectric transducers. Acoustic streaming leads to effective damage-free cleaning. Originally developed at RCA, several types of units are available from commercial equipment suppliers. Combined with diluted SC-1 solution at 70°C megasonics affords very efficient simultaneous removal of submicron particles and chemical contaminants from the front and back surface of wafers.

Single-wafer brush scrubbers hydrodynamically remove particles of size 1 μm and larger from the front and back side of hydrophilic FEOL and BEOL wafers. Special brushes and optimized cleaning solutions, UPW, or IPA are used. Brush scrubbers have been used since the early days of wafer processing and in their refined version are still a choice piece of equipment for particle removal after CMP.

Wafer carriers are an integral part of all wafer processing operations because of the need of handling and transfer of wafers in batches. Wafer carriers are designed to hold wafers in slots of 25 or more wafers. A variety of plastic materials are available for their manufacturing. They can be a critical source of particle contamination, their proper use and specialized cleaning are very important for preserving wafer cleanliness. For a detailed discussion of this important topic, which is beyond the scope of this chapter, the reader should consult the specialized literature.

1.4.5 Wafer Rinsing, Drying, and Storing

The last steps in wet-chemical wafer cleaning are rinsing and drying [176,177]; both are extremely critical because clean wafers can become recontaminated very easily. Rinsing should be done with flowing high-purity and ultrafiltered high-resistivity DIW at room temperature. The results of several early studies were published some years ago [37,178−180]. More recent papers include fundamental/theoretical aspects of rinsing [165,181−184], optimization for single-wafer rinsing [182] and by acidification to prevent metal deposition [52,165,185], corrosion prevention in BEOL rinse processes [186], and physical mechanisms involved in rinsing patterned wafers [182−184].

1.4.5.1 Wafer Rinsing

Megasonic rinsing is advantageous [179] and is the most effective technique for reducing the critical boundary layer between the wafer surface and the rinse H_2O [37,178−180]. BEOL wafers with Cu metallization can be safely rinsed megasonically at 60°C [187].

Centrifugal spray rinsing [108,188] and processing in a *closed-system module* [109,189] have the advantage that the wafers remain enclosed between the cleaning, rinsing, and drying steps.

Quick dump rinse has been the most frequently used method and is still being used widely.

1.4.5.2 Wafer Drying

Wafer drying after rinsing must be done by physical removal of the H_2O, rather than by allowing it to evaporate, which would leave residues.

Spin drying accomplishes this and has been one of the most widely used techniques, although recontamination tends to occur.

Forced air or N_2 drying using warm and filtered gas is a preferred technique with less chance for particle recontamination [190,191].

Capillary drying is based on capillary action and surface tension to expel the H_2O. Individual wafers in a plastic holder are slowly pulled out of DIW at 80−85°C into a controlled atmosphere. Less than 1 wt% of the H_2O is claimed to evaporate, leaving a particle-free surface [176].

Solvent vapor drying starts with wet wafers being moved into the hot vapor of a pure, H_2O-miscible solvent, usually IPA, which condenses and thereby displaces the H_2O. The wafers dry particle-free when the cassette is withdrawn above the vapor zone. IPA vapor drying in its various modifications has become the preferred method in batch processing. Variously operating commercial drying systems for IPA (and for nonflammable solvent mixtures) are available [109,176,192]. The purity of the solvent is extremely important, and the H_2O content during processing must be closely controlled so as not to exceed a critical concentration to achieve an ultraclean and dry surface [109,176,193−195]. A comparative evaluation of spin rinse/drying and IPA vapor drying is available [196]. Secondary ion mass spectroscopy (SIMS, discussed in Chapter 10) could not detect IPA on Si surfaces, but the growth rate of native SiO_2 films was depressed, indicating the presence of a very thin IPA film. Interestingly, the electrical properties of the oxide improved substantially [197].

Marangoni drying is a different type of vapor drying. During a slow removal of the wafer from the rinse UPW, the air/H_2O/Si surface is exposed to a H_2O-miscible tensioactive organic solvent vapor, usually IPA. Surface tension effects cause the H_2O to sheer off a planar wafer surface [198], leaving a clean and dry hydrophilic Si surface. Marangoni IPA wafer drying techniques have been incorporated widely in commercial wet-cleaning systems.

Rotagoni, a single-wafer drying method, combines the Marangoni drying principle with a single-wafer spin machine. It can be used in a low rotational speed, which can significantly reduce the amount of splash back and entrainment of airborne particles. After spin rinsing with UPW, spin drying is done in an atmosphere of IPA [182,199].

1.4.5.3 Wafer Storage

Extreme care must be taken to avoid recontamination of clean device wafers during storage if immediate continuation of processing is not possible. Wafers should be placed ideally in chemically cleaned, closed glass containers or stainless steel enclosures, while being flushed with high-purity filtered N_2 in a clean room. However, this is rarely feasible in the fab; the usually used plastic containers will always cause organic surface contamination. Metal tweezers must never be used to handle semiconductor wafers because they will invariably cause contamination by traces of metals. The final criteria of the success of all wafer cleaning operations are the purity of the wafer surface after the last treatment and eventually the semiconductor device yield. No matter how effective the various cleaning steps may be, improper rinsing, drying, and storing can ruin the best results [105].

1.4.6 Dry Cleaning and Surface Conditioning Processes

1.4.6.1 General Considerations

Dry cleaning as opposed to wet cleaning is the subject of Part III of this book and covers three chapters:

1. Vapor-phase cleaning
2. Plasma stripping and cleaning
3. Cryogenic aerosol/supercritical fluid cleaning.

The two chapters on plasma and cryogenic cleaning are entirely new in this second edition of the book and have been added because of the increased importance of these technologies. The dry cleaning methods and processes to be overviewed should not necessarily be taken as an outright replacement of wet-cleaning, but should rather be considered a valuable complement in certain process sequences to achieve optimal results. Dry cleaning methods are presented in Part III, Chapters 5–7. Early insights of this cleaning method are included.

1.4.6.2 Vapor-Phase Cleaning Processes and Methods

Vapor-phase cleaning is often called *dry-cleaning* in contrast to *wet-cleaning*. However, just like liquid cleaning is a more accurate term than the more restricted "wet-cleaning," vapor-phase cleaning is a more concise expression than "dry-cleaning" because it is done in the gas or vapor phase, which may or may not be dry. Vapor-phase cleaning offers actual and potential advantages over liquid cleaning methods in the fabrication of advanced semiconductor devices in terms of processing and economy. In this section, we present a preview of vapor-phase cleaning processes for the removal of various types of contaminants and for conditioning the Si surface. An in-depth presentation will be presented in Chapter 5.

The impetus for developing vapor-phase cleaning processes has been to correct shortcomings of wet-chemical processing, such as ineffective removal of cleaning solutions from submicron device structures with deep and narrow trenches, incompatibility with process integration, and high cost of the large quantities of liquid chemicals and UPW required. Vapor-phase cleaning can correct these problems and allow process integration for cluster tooling at reduced pressure. However, particles cannot be readily removed from the surface, the reaction of adsorbed chemical impurities is generally specific rather than all-inclusive, and H_2O rinsing may still be required to remove reaction products.

Four major processes have emerged for practical applications of vapor-phase cleaning in IC device fabrication:

1. HF vapor etching for removal of oxides
2. UV/O_3 exposure for removal of organics
3. UV/Cl_2 processing for removal of adsorbed metal contaminants
4. Organochemical vapor-phase cleaning for metal removal under less aggressive conditions.

1.4.6.2.1 HF Vapor Etching, Cleaning, and Surface Conditioning

Films of SiO_2 (native, deposited, and thermal) and silicate glasses (phosphosilicate glass; borosilicate glass; and borophosphosilicate glass) can be vapor etched with anhydrous HF gas, HF gas/H_2O vapor mixtures, and HF gas/alcohol vapor, all with N_2 carrier gas. A variety of mixtures, pressure conditions, temperatures, and reaction times have been used in practice. One method for attaining good control and uniformity of the etching process is based on purging the reactor with dry N_2 followed by introducing anhydrous HF gas without additives. After an initiation step by condensation of HF and H_2O, the subsequent etching proceeds by the same chemical reaction as in liquid phase wet etching, even though no H_2O vapor is intentionally introduced. Small quantities of H_2O or alcohol present in the ambient air or adsorbed on the wafer surface and reactor walls are sufficient to start the etching reaction. The H_2O resulting from the HF/SiO_2 etching reaction, as a byproduct, is adequate to sustain continued etching. A slightly elevated temperature (40°C) is maintained at a sufficiently high gas flow rate to remove reaction products, usually at subatmospheric pressure (i.e., 100 torr). An excellent description of the detailed HF vapor etching mechanism originally outlined by B. E. Deal and C. R. Helms is partially incorporated in Chapter 5.

The etching rate of SiO_2 layers for thickness reduction can be precisely controlled by adjusting the HF gas flow rate or the temperature to achieve very good uniformity of the residual oxide thickness over the entire wafer surface. Both IPA and methanol (CH_3OH) have been used as reaction additives to anhydrous HF gas. There is evidence that some adsorbed metal contaminants are also eliminated with the oxide from the Si surface. Complete removal of oxide layers down to the Si results in a hydrophobic (hydrogenated) surface. The process is an excellent prethermal, preconduct, preepi, and predeposition clean for surface conditioning FEOL wafers. It has also been used for in situ cleaning of conduct areas before metal deposition. Reactor systems for commercial processing are available, such as the original Excalibur, manufactured by FSI [200].

Specifically, removal of native oxides from Si surfaces using anhydrous HF at low temperature before polySi deposition has been described [201]. Polymeric/silicate residues, SiO_2, and metal oxides in via holes can be removed with anhydrous HF [202]. Anhydrous HF gas with IPA vapor and N_2 for SiO_2 etching at 150 torr and 50°C has been described [203] and HF gas with CH_3OH vapor additive [204,205]. Oxide removal with HF/H_2O vapor at atmospheric pressure has been used for CMOS (complementary metal oxide semiconductor) production [206], for deep submicron n-MOSFET (MOS field effect transistor) cleaning at low pressure [207], and for cluster tool processing [208]. Etching reactions of thermal SiO_2 and silicate glass films with HF−H_2O vapor at low temperature (40°C) and reduced pressure (250 torr) were studied using in situ Fourier transfer infrared analysis [209]. Finally, selective etching of native oxide films can be accomplished with azeotropic HF−H_2O and anhydrous HF−IPA [210].

1.4.6.2.2 Ultraviolet/Ozone Cleaning for Removal of Organics

Short-wave ultraviolet (UV) radiation in the presence of O_2 is the oldest of the vapor-phase cleaning methods that has been used successfully for removing organic surface contaminants from many types of surfaces. It is essentially a photosensitized oxidation process. Briefly, many types of organic molecules absorb the short-wavelength UV radiation of 184.9 nm that is generated by low-pressure Hg discharge lamps; this wavelength is also absorbed by the O_2 in the ambient air, generating O_3. The coproduced 253.7-nm radiation is not absorbed by O_2, and hence does not contribute to O_3 formation, but it is absorbed by most organic molecules and also by O_3. This absorption by O_3 is primarily responsible for its destruction so that O_3 is continually being formed and destroyed. Atomic oxygen is generated as an intermediate product and its strong oxidation potential allows the reaction with excited or dissociated contaminant molecules, free radicals, and ions. It is probably the main reason for the effectiveness of UV/O_3 cleaning. The combination of short-wavelength UV and O_3 has a much greater cleaning efficiency that UV light alone or O_3 alone. The reaction products consist mainly of H_2O, CO_2, and N_2 and may contain volatile organic degradation products.

Primary applications of UV/O_3 cleaning and surface conditioning are for stripping of photoresists and polymer residues and for removal of organic and carbonaceous contaminants to improve photoresist adhesion, typically at 120°C and 500 torr [211]. A discussion of UV/O_3 cleaning and its evolution by J. R. Vig is partially included in Chapter 5.

1.4.6.2.3 UV/Chlorine Vapor-Phase Cleaning for Metal Removal

Vapor-phase removal of metal contaminants can be achieved by reaction with UV-excited chlorine. The photo-excited Cl species may etch a thin (i.e., 30 Å) Si layer, removing Cu and other trace metal impurities that may be present in or on surface-damaged Si layers by forming volatile metal chloride complexes [205,209,212]. The reaction conditions must be carefully optimized to avoid pitting and to produce a smooth Si surface. Temperatures from about 50 to 400°C have been used under low-pressure conditions with H_2 as a diluent. Metals that have been successfully reacted and removed include Cu, Fe, Cr, and Ni; organic surface contaminants may be eliminated at the same time [213]. The performance of nanometer-thin gate oxides grown on smooth UV/Cl_2-treated Si surfaces was found to be superior [214]. UV reactor modules for FEOL single-wafer processing are now available commercially [214]. UV/Cl_2 processing has been used for critical pregate oxidation Si cleaning in conjunction with anhydrous HF/IPA/H_2O for oxide removal, using a vacuum clustered reactor system [215].

1.4.6.2.4 Organochemical Vapor-Phase Cleaning Processes

Approaches based on less aggressive chemical reactions with organic chemicals are aimed at the formation of volatilizable compounds, such as metal chelates or nitrosyl compounds. The key requirement for removing trace metal impurities is the formation of volatilizable species by reaction at low temperature to prevent diffusion, followed by their elimination at an elevated temperature at low pressure. Special attention must be paid to the state in which contaminants occur. Elemental metals and other impurities are often present as absorbates or inclusions in the native oxide film rather than being exposed on the semiconductor or oxide surfaces. Etching in HF gas may be necessary to first remove the oxide envelope to render the impurities accessible for chemical attack.

The removal of particles and particulates from wafer surfaces is another difficult problem. A chemical reaction could possibly transform them into volatile species, depending on their composition and size. Vapor etching of an oxide film on which particles are located or in which they are imbedded can be effective if the wafers are positioned vertically and if a vigorous stream of inert gas is applied to sweep the freed particles off the wafer surface.

One of the most frequently used chelating reactant is 1,1,1,5,5,5-hexafluoro-2,4-pentanedione, which is used to form volatile metal coordination compounds of Cu [216,217], Fe [217,218], and Na [219]. Copper and iron contaminants are usually present in the form of their various oxides or hydroxides [216–218]. The removal of submonolayer quantities of Cu, Fe, and Na and Ca, K, Li, Ni, and Zn has been studied in detail [220]. Reaction conditions vary widely for different species of impurities and so do the effectiveness for their removal or for the lowering of their surface concentrations, limiting commercial applications for semiconductor device fabrication.

1.4.6.3 Plasma Stripping and Cleaning

1.4.6.3.1 Introduction

The manufacturing of ICs is a repetitive batch process which includes the application of a patterned photopolymer mask to block ions from being implanted or to prevent etchants from reacting with selected areas of the substrate. Deposited layers of insulators, polySi dielectrics, metals, diffusion barriers, and surface passivation protects—all need to be patterned. The photolithographic patterning process invariably consists of selective photoresist masking followed by ion implantation or by etching of the exposed areas and finally the stripping of the bulk photoresist mask that has served its purpose. Stripping can be accomplished with various degrees of effectiveness using organic solvent mixtures, liquid chemicals, or dry techniques such as UV/O_3 or exposure to a plasma environment. Plasma-enhanced stripping, also known as "plasma ashing," is the method of choice because of its versatility and efficiency. Plasma stripping has been considered part of the specialized photolithographic processing rather than of wafer cleaning. However, the additional applications for removing etch residues from vertical profiles and from surface cleaning and conditioning have made plasma-enhanced processing an important part of wafer cleaning. Consequently, we have added

Chapter 6 in the second edition of this book, which discusses the subject in considerable depth. The important technology of remote plasma processing is presented in Chapter 6 and reviews of these technologies originally authored by R. A. Rudder, R. E. Thomas, R. J. Nemanich, and J. Ruzyllo are included.

The major applications of plasma-assisted stripping include

1. Stripping of bulk photoresist material
2. Removing etch residues
3. Predeposition cleaning
4. Surface conditioning.

These topics will be briefly addressed as an introduction to Chapter 6.

1.4.6.3.2 Stripping of Bulk Photoresist

Photoresists are essentially hydrocarbon polymers composed of a novolack resin, a photoactive compound and an organic solvent. The removal of bulk photoresist patterns is typically accomplished by reaction with atomic oxygen (O), which is created in the plasma environment by dissociation of molecular O_2 [221,222]. Basically, two types of plasma reactors can be used to conduct the process both designed for single-wafer processing.

1. A design based on remote or downstream plasma generation
2. A design based on the combination of remote with RF, microwave, or ECR sources for downstream plasma generation with RF-assisted bias.

The preferred method of bulk photoresist stripping after plasma pattern etching or ion implantation now uses downstream reactors, which minimizes ion-induced surface damage while providing good control over reaction parameters with O_2 [223]. The stripping rate can often be increased by adding specific gases or vapors to the O_2 plasma such as F-containing gases [224], or H_2O vapor, which provides additional benefits [225]. Device structures, which cannot withstand oxidation require a nonoxidizing environment. A mixture of typically 10 vol% H_2—90 vol% N_2 can then be used as the reactive gas in the plasma cleaning procedure [201].

Patterned photoresist is often used as a mask for the selective implantation of ions. During this process, the surface of the photoresist becomes hardened, making it difficult to remove the used resist layer. Low-energy treatment in an O_2 plasma containing an optimally controlled quantity of H_2O vapor can remove the resist without damage to the substrate [226].

1.4.6.3.3 Removal of Etch Residues

Plasma etching used for generating high—aspect ratio Si trenches requires the formation of an etching-resistant sidewall passivation layer to provide protection from lateral etching [227]. Plasma-assisted cleaning is often not adequate to substantially remove these postetch passivation layers, which may consist of polymers, inorganic materials, etchants, and oxide or metal residues. More reactive alternatives to O_2 are necessary such as NO, N_2O, CF_4, or NF_3. The addition of a small amount of atomic F to the atomic O present in the downstream microwave discharge followed by a UPW rinse can be very effective [228].

The removal of metal etch residues can be especially difficult because of Cl contamination, which originates from the metal etching process. Chlorine may become absorbed in the photoresist mask and the sidewall passivation layer and pose dangerous corrosion problems [222]. The application of in situ H_2O-based downstream plasmas followed by ex situ wet-chemical process steps can remove these corrosive contaminants, photoresist residues, and sidewall polymers [229]. It should be mentioned that nearly all plasma-cleaned wafers require a follow-up with wet-cleaning as described in this chapter and Chapter 4 to remove residual impurities and reaction products.

1.4.6.3.4 Predeposition Cleaning

Pregate dielectric cleaning and surface conditioning are especially critical process steps that require minimizing radiation damage, surface recontamination, and roughening of the Si surface. The quality of the thin gate oxide deposited subsequently is directly related to the physical condition of the substrate surface.

A second example of a predeposition treatment is plasma-assisted cleaning before epitaxial growth of Si. It is done primarily by remote plasma processing in the reducing atmosphere of H_2. The plasma generates active H, which can interact with many types of surface contaminants and effect their destruction and removal.

1.4.6.3.5 Surface Conditioning

The chemical and physical conditions of the Si substrate and deposit surfaces can be modified by reactive plasma reactions to optimize for specific process steps. This surface conditioning may be the result of, or can be combined with, the plasma cleaning procedure. For example, in situ cleaning and surface conditioning of the Si surface to produce smooth and atomically clean surfaces with controlled Si H-termination bonds immediately before Si epitaxy [230] can be achieved by activated hydrogen plasma processing in a remote plasma reactor. Remote hydrogen-plasma cleaning in situ has also been applied successfully to wafers with patterned $Si{-}SiO_2$ surfaces [231].

1.4.6.3.6 Cryogenic Aerosol and Supercritical Fluid Cleaning

The emerging technologies of cryogenic and supercritical cleaning have become important wafer cleaning methods in areas where conventional processes are no longer adequate. An excellent example is the cleaning and surface conditioning of BEOL wafers with low-k films and other sensitive materials or fragile structures that cannot withstand liquid chemical or conventional dry cleaning. Therefore, new Chapter 7 has been added to address this important new enabling technology and its status to date. The notes that follow may serve as a brief introduction to that chapter.

1.4.6.3.7 Cryogenic Aerosol Cleaning and Conditioning

Cryogenic aerosols are submicron colloidal crystal particles of a cryogenic gas, which are dispersed in the gas phase. Initially, solid aerosol particles are formed when the temperature of a cryogenically cooled and compressed gas drops below the so-called triple point, the location in the temperature–pressure phase diagram where solid, liquid, and vapor coexist in equilibrium. The aerosol is generated when the cryogenically expansion-cooled mixture is dispensed over a wafer surface from a cooled distributor nozzle. Cryogenic cleaning with diluted argon in nitrogen (Ar/N_2) is conducted at subatmospheric pressure. In the case of CO_2, cleaning is done at atmospheric pressure; the dense white spray that results is known as "CO2-snow" (SCO_2). Cryogenic wafer cleaning is highly directional so that removal of contaminants occurs in line-of-sight. Removal of organic contaminants requires a liquid phase, which is chemically inactive, whereas particle removal involves strictly kinetic interactions of aerosol crystals with contaminant particles.

A theoretical analysis of wafer cleaning with cryogenic aerosols (Ar/N_2) has been published [232]. Wafer cleaning with cryogenic Ar/N_2 aerosols is especially useful for IC device wafers with metallization and sensitive interconnect structures because cleaning is nonreactive, noncorrosive, and nondamaging and can substantially reduce the number of defects [233,234]. Cryogenic (or cryokinetic) cleaning has also been used for efficiently cleaning BEOL wafers with Cu metallization and low-k materials without introducing any damage or chemical changes in the materials [235]. Cryogenic aerosol processors for cleaning Si device wafers in the fab with Ar/N_2 are available commercially [233,234].

The formation of SCO_2 follows the same physical principles as in the generation of Ar/N_2 aerosols, but the technical requirements for its implementation are simpler and less demanding. SCO_2 is formed by passing liquid and/or gaseous CO_2 through a small-aperture nozzle. The mixture rapidly expands and cools, resulting in lowered pressure/temperature with the consequent nucleation of solid CO_2 particles entrained as an aerosol in CO_2 gas. The exiting high-velocity stream leads to momentum transfer from the SCO_2 to the solid contaminant particles, which are dislodged and removed with the gas stream. Organic impurities are probably removed by a liquid CO_2 phase [236–239]. SCO_2 cleaning has been shown to remove submicron particles and residues from post-CMP surfaces more completely than brush scrubbing and other liquid and gaseous cleaning processes [240,241]. Commercial processing systems with refined nozzle design for wafer cleaning with SCO_2 have become available [237,239].

1.4.6.3.8 Supercritical Fluid Cleaning

Supercritical fluids have been used extensively in extraction and purification processes in many industries. The advantages in using supercritical fluids center in their unique nature; these fluids feature low viscosity and negligible degrees of surface tension, which allows gaslike diffusivity and high penetration capabilities and their density is sufficiently high to match the solvating properties of a liquid.

The supercritical liquid of choice for Si wafer cleaning and surface conditioning is carbon dioxide. CO_2 reaches the supercritical state above its critical temperature of 31°C and the pressure of 75 bar. The liquid and vapor phases of CO_2 then merge forming supercritical CO_2 ($scCO_2$). In this state the supercritical fluid combines the best aspects of both wet and dry cleaning agents, featuring the higher density of liquids and the lower viscosity and very low surface tension of gases. It can therefore penetrate, clean, and dry IC device features with geometries in the nanometer range. It is compatible with nonporous and porous low-k dielectrics and can effectively remove residues and submicron particles [237,238,242−244]. Small amounts of surfactants, chelating agents, and cosolvents added to $scCO_2$ can further enhance the remarkable cleaning effectiveness [242−244]. It is obvious that this new emerging technology can offer viable solutions to critical problems in BEOL single-wafer cleaning where wet-chemical processing has severe limitations, in low-k resist stripping and sidewall polymer removal, damage-free elimination of submicron particles, and cleaning of device structures with high−aspect ratio geometries. A considerable number of papers have been published recently on applications of $scCO_2$ cleaning in microelectronic device manufacturing, as noted in Refs. [242,245−247].

1.5 EVOLUTION OF WAFER CLEANING SCIENCE AND TECHNOLOGY

In this section, we will trace the evolution of semiconductor wafer cleaning science and technology from the advent of semiconductor device fabrication to the present time. Five periods of this development can be discerned:

1. The early days from 1950 to 1960
2. Development of wet-chemical cleans from 1961 to 1971
3. Evaluation, application, and refinement of wet cleans from 1972 to 1989
4. Era of explosive research and development activity for both wet and dry cleaning methods from 1989 to 1992; and finally
5. Era of modern cleaning science and technology from 1993 to the present.

The introduction section of each chapter in this book usually includes historical remarks pertaining to the chapter topic. The present systematic survey of the entire field is intended to provide a comprehensive perspective of this evolution. The limited and manageable amount of information published during the developing periods up to 1993 allows for a detailed review, identifying specific researchers. The overwhelming quantity of published material during the modern era to the present necessitates a limited survey of only selected representative key advances.

1.5.1 Period from 1950 to 1960

Harmful effects of impurities on the performance of simple transistors were recognized already in the early days of Ge (germanium) processing, but became more apparent with the advent of Si transistor fabrication in the later 1950s. Some sort of wafer cleaning was deemed necessary as part of the device manufacturing process. Early cleaning techniques consisted of mechanical and chemical treatments. Particulate impurities were removed by ultrasonic treatment in detergent solutions or by brush scrubbing. The first caused frequent wafer breakage and the second often deposited more debris from the bristles than impurities it removed from the wafer surface. Organic solvents were used to dissolve wax residues and other soluble gross organic impurities and photoresists.

Chemical treatments for cleaning consisted of immersion of the wafers in aqua regia (1 vol:3 vol HNO_3:HCl), concentrated HF, boiling HNO_3, and mixtures of hot acids. Combinations of H_2SO_4 and chromic acid (H_2CrO_4 and $H_2Cr_2O_7$ based on CrO_3) led to chromium contamination and caused ecological toxicity problems on disposal. Mixtures of hot H_2SO_4 and H_2O_2 caused S contamination. Nitric acid and HF were impure to begin with and led to redeposition of impurities, such as heavy metals including traces of Au. In general, impurity concentrations and particle densities in process chemicals were high and in themselves tended to lead to surface contamination. Aqueous solutions containing H_2O_2 had long been used for cleaning electron tube components [248,249], but were never investigated for possible application to semiconductor wafer cleaning. Plasma ashing was the first dry process applied to wafer cleaning for removing organic photoresist patterns but left high concentrations of metallic compounds and other inorganic impurity residues.

1.5.2 Period from 1961 to 1971

This period can be considered one of research into semiconductor surface contamination and the systematic development of wafer cleaning procedures. The need for effective wafer cleaning became critically apparent during this period when Si-ICs were beginning to move into large-scale production. Although the IC was invented in 1958 (by Jack Kilby from TI and Bob Noyce from Fairchild), it took several years before production actually started.

1.5.2.1 Radiochemical Studies of Surface Contamination

Radioactive isotopes (radionuclides) offer a unique opportunity for the study of surface contamination with an unprecedented degree of sensitivity. Only a few papers had been published before 1963 on the use of radioactive tracers for investigating surface contamination, notably, one by Wolsky et al. for Ge [250] and papers by Sotnikov and Belanovskii [251] and by Larrabee for compound semiconductors [252]. In a series of papers on extensive contamination studies, Kern applied radioactive tracer methods to investigate the concentrations of contaminant elements that had transferred onto electronic materials during manufacturing operations. It may be of interest to look at some of these still useful early results, which were published in 1963 [253,254].

The adsorption of Na ions on the assembly parts of Ge transistors (RCA 2N217npn) during anodic etching of the Ge tabs in alkaline electrolyte solution was investigated with radioactive Na-22 as tracer. The efficiency of various rinsing treatments for desorbing Na adsorbates was assessed by measuring the radioactivity as a function of treatment time. Germanium pellets that had been cascade-rinsed with DIW had a concentration of 6.2×10^{14} Na atoms/cm^2. The plots in Fig. 1.5-7 demonstrate several interesting results.

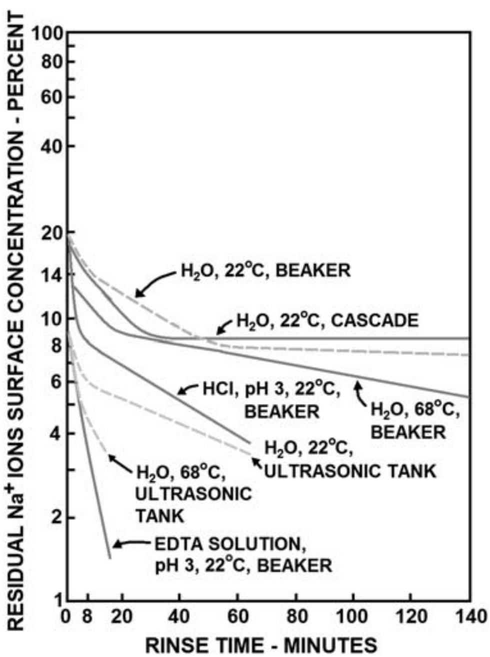

FIGURE 1.5-7 Efficiency of various desorption treatments for removing Na adsorbates from Ge transistor assemblies. The transistors had been electrolytically etched in NaOH solution containing radioactive Na-22 ions as the tracer [253]. *Used with permission from Solid State Technology.*

For example, EDTA chelating solution was 280 times more effective than countercurrent cascade rinsing with cold UPW. A similar investigation was conducted with the components of a Si power transistor (RCA 2N1482pnp). In this work Na-24 was created by thermal neutron activation to attain several thousand times greater radioactivity levels than were available for the work with the germanium transistors, achieving a much greater analytical sensitivity. The final concentration on Si wafers after acidic desorption was less than 8×10^{11} Na atoms, or nearly 1/1000th of one monoatomic layer/cm^2. Contamination of Si transistors by metallic impurities from solutions was assessed by use of the radioactive isotopes Cr-51, Fe-59, Cu-64, and Au-198. Techniques were devised to minimize residual concentrations leading to a production yield increase for a Si power transistor (RCA 2N2102) of over 200%.

The contamination of wafers of Si, Ge, and GaAs during wet-chemical etching and processing was investigated by adding known quantities of radioactive trace metals to various processing solutions. The spatial distribution of residual metals after rinsing with UPW was examined by autoradiographic film techniques and the average surface concentrations were determined by correlation with radiation intensity measurements. The effectiveness of various rinsing and cleaning treatments was also measured quantitatively. The results of this work were published in 1963, including print copies of autoradiograms [253,254].

Similar but more detailed studies were conducted later on and reported in a series of five papers in 1970 (255 I–III) and 1972 (256 I, II). The adsorption and desorption of etchant *constituents* on semiconductor wafers were investigated by use of the radionuclides Na-22 and Na-24 for NaOH, F-18 for HF-containing etchants, Cl-38 for HCl, I-131 for polishing etchants, and C-14–labeled acetic acid (CH_3COOH) for etchant mixtures for Si. An interesting example of the desorption behavior of Na adsorbed on Si is presented in Fig. 1.5-8 showing the effect of aging on diminishing desorption efficiency, which is caused by entrapment of Na in the growing oxide layer. The wafers had been immersed in 0.025 N NaOH tagged with Na-22 followed immediately by initial rinsing for 60 s in UPW at 20°C. Subsequent desorption treatments within 24 h (open symbols) are compared with those after several weeks of storage (solid symbols) [255].

Surface contamination by trace impurities from various etchants and cleaning solutions was investigated with radioactive Mn-54, Zn-65, Mo-99, Sb-122, and Sb-124 in addition, to the previously used metal tracers Cr-51, Fe-59, Cu-64, Au-168, plus the reagent *constituent* tracers C-14, F-18, Na-22, Na-24, Cl-38, and I-131. Disks of fused quartz (SiO_2) were used as substrates in addition to wafers of Si, Ge, and GaAs. The adsorption of contaminants was measured for many typical etchants and reagents containing the radioactively marked ions; the desorption efficiency was determined for DIW and many chelating and cleaning solutions. For example, acidic H_2O_2

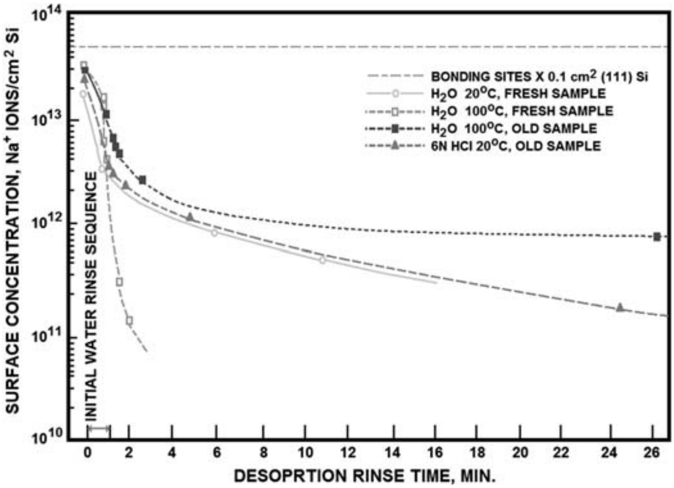

FIGURE 1.5-8 Inhibition effects of aging on desorption of Na ions from Si wafers with DIW and with 6 N HCl [255]. *Used with permission from RCA Review.*

solutions were most effective for desorbing Au, Cu, and Cr from Si and Ge, and HCl for Fe on Si. EDTA and other chelates suppressed deposition of Cu on Ge from solutions by two to three orders of magnitude.

A few typical metal adsorption and desorption plots for Si are reproduced in Figs. 1.5-9–1.5-15 and for silica in Fig. 1.5-16. The strong concentrating efficiency of Si for Au from HF solution is shown in Fig. 1.5-17 and its possible utilization for purifying HF solution is demonstrated in Fig. 1.5-18.

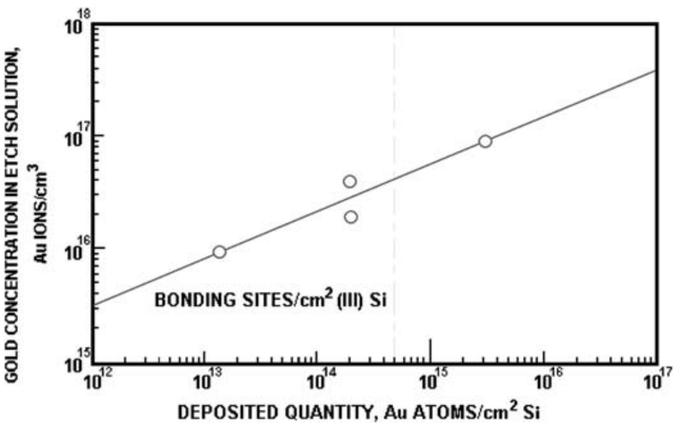

FIGURE 1.5-9 Gold atoms deposited from an etchant on Si wafers as a function of Au ion concentration in solution. Au-198 was added as the radioactive tracer to the HF−HNO_3−CH_3CO_2H−I_2 etchant containing Au ions [255]. *Used with permission from RCA Review.*

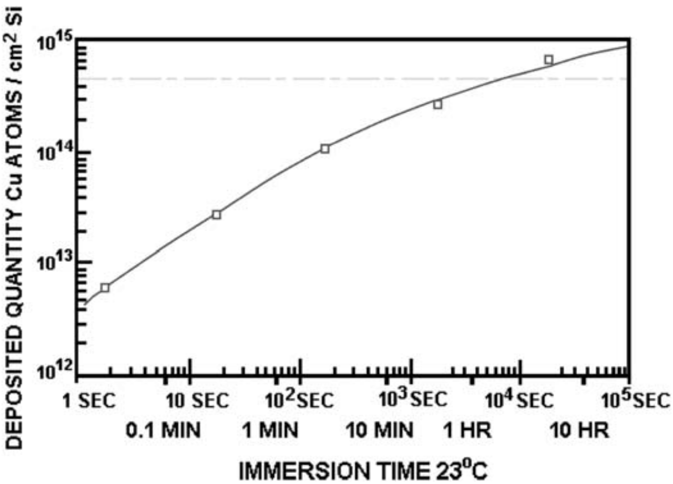

FIGURE 1.5-10 Copper atoms deposited from a 49% HF solution on Si wafers as a function of immersion time at 23°C [255]. The acid contained 0.1 ppm of Cu as cupric ions tagged with Cu-64. The *dashed line* represents the number of available bonding sites per cm^2 of Si <100>. *Used with permission from RCA Review.*

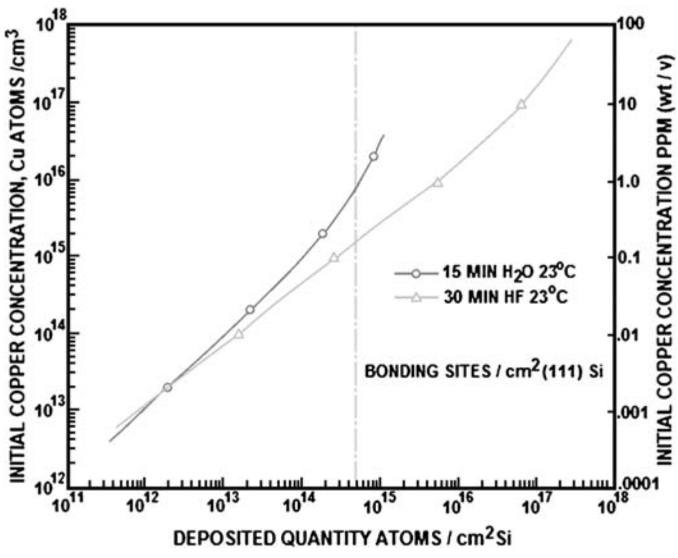

FIGURE 1.5-11 Quantity of Cu deposited on Si wafers from a 49% HF solution and from DIW as a function of copper concentration in solution. The solution contained 0.1 ppm Cu as cupric ions tagged with Cu-64. The *dashed line* represents the number of available bonding sites per cm^2 of Si <111>. Before immersion in the radioactive water, the wafers had been dipped in nonreactive HF and rinsed in nonradioactive H_2O [255]. *Used with permission from RCA Review.*

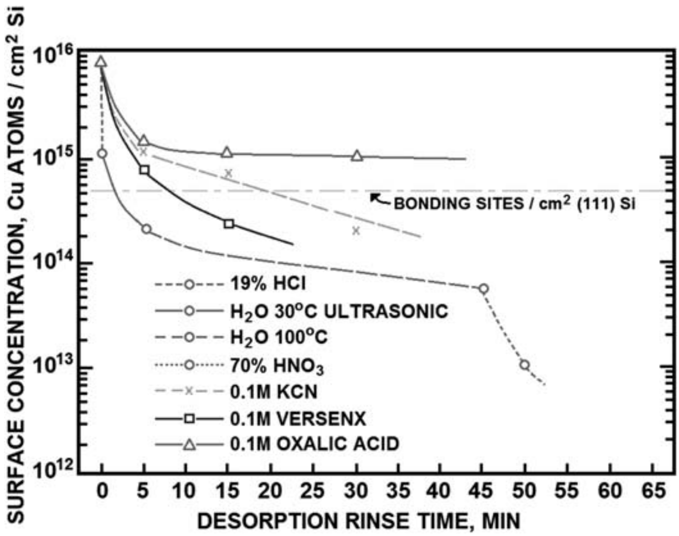

FIGURE 1.5-12 Effectiveness of various cleaning agents for desorbing thick Cu films deposited from hot 5% NaOH with Cu-64 tagged ions. Desorbing treatments were conducted at 23°C except for the water rinses. The chelating agent (VERSENX 80) was pentasodium diethylenetriamine pentacetate [255]. *Used with permission from RCA Review.*

The dashed gamma radiation spectrum in the first figure shows that a 1 N HF solution containing Sb-122, Mo-99 with its associated Tc-99, and Au-198 presents only as a minor peak. The solid curve obtained from Si wafers that had been immersed in this solution was normalized with the Tc-99m peak

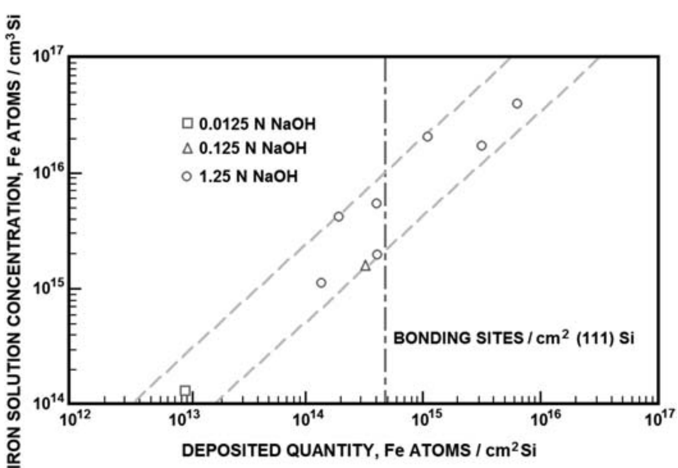

FIGURE 1.5-13 Quantity of Fe deposited from various NaOH solutions on Si wafers as a function of Fe solution concentration. Fe-59 was used as the radioactive tracer. The wafers were immersed at 100°C for 1 min followed by a 30-s prerinse [255]. *Used with permission from RCA Review.*

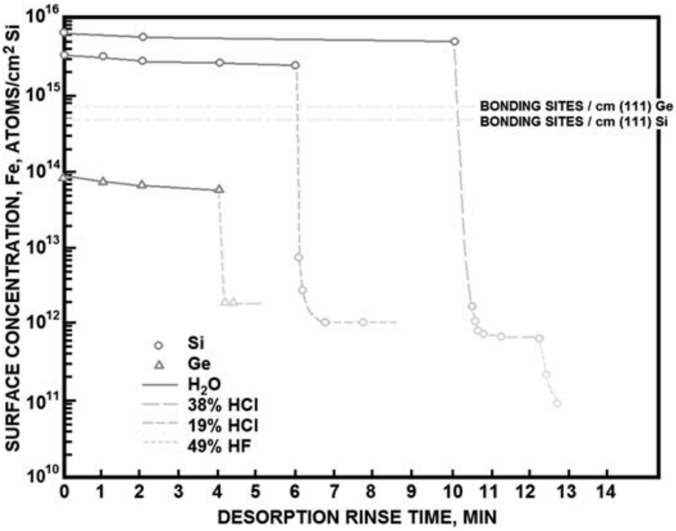

FIGURE 1.5-14 Efficiency of DIW and acid solutions at 23°C for desorbing Fe deposits from Si and Ge wafers. Fe-59-containing deposits from hot NaOH were used. Surface quantities were decreased up to five orders of magnitude to the detection limit [255]. *Used with permission from RCA Review.*

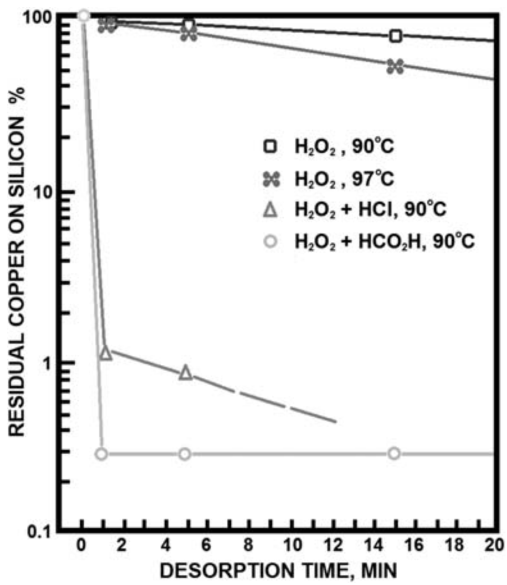

FIGURE 1.5-15 Desorption efficiency of various agents for removing thick Cu-64-tagged deposits from Si wafers. The initial Cu concentration from HF solution was five times higher for the H_2O_2—HCl test than for the others [101]. *Used with permission from RCA Review.*

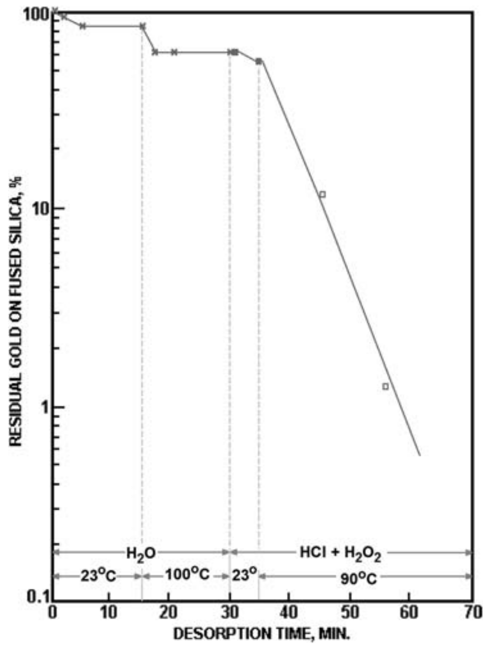

FIGURE 1.5-16 Desorption efficiency for Au from fused quartz surfaces under various conditions. The quartz plates had been etched in 49% HF solution containing Au ions with Au-198. The initial surface concentration was 1.6×10^{12} Au atoms/cm^2. The H_2O:H_2O_2:HCl mixture consisted of 8 vol H_2O, 1 vol HCl (1N), and 1 vol H_2O_2 (30 wt%) [101]. *Used with permission from RCA Review.*

maximum of the solution spectrum. The extremely high degree of selective deposition of metallic Au on the Si is dramatically evident by comparing the two Au-198 radiation peak intensities [255,256]. The utilization of this effect is exemplified in Fig. 1.5-18; percolating a 49% HF solution containing Au-198 through a column of high-purity Si crystal pieces resulted in retention of more than 98% of the Au in the top sixth of the column and more than 98.8% removal in a single pass. Similar results were obtained by Cu and other heavy metals in HF, BHF, and H_2O_2 solutions [253].

A unique approach for studying contaminant transfer by radiochemical techniques concerns the transfer of impurity elements at high temperature from crucibles to semiconductor ingots [257]. Fused quartz (SiO_2) crucibles were radioactivated by bombardment with thermal neutrons in a nuclear reactor to produce radioactive Si-31 from the quartz and various radionuclides from its impurity elements. Crystals of GaAs were then grown in these highly radioactive crucibles. After cooling, gamma ray scintillation spectrometry of ingot sections allowed identification and quantitative decay measurements of the transferred contaminants. Radioactive Si-31, decaying by gamma ray

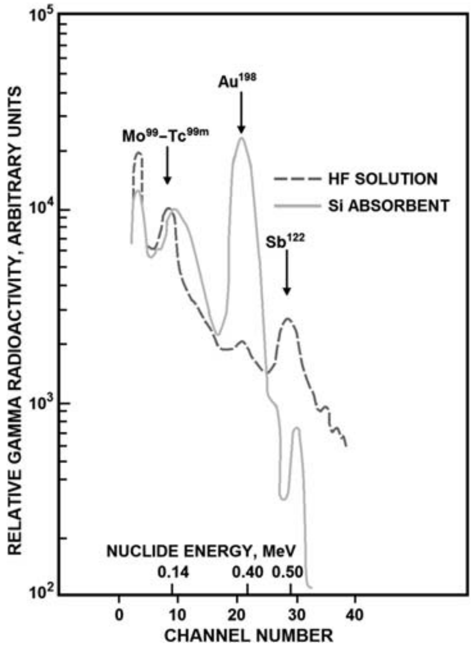

FIGURE 1.5-17 Gamma radiation spectra of a 1 N HF solution and its adsorbate on Si. The relative distribution of Au, Mo, and Sb in the solution and in the resulting adsorbate demonstrates the extreme concentration effectiveness of Si or Au [255]. *Used with permission from RCA Review.*

emission to P-31, was the main radiation product utilized in this study. The results led to improved processing techniques for the synthesis of GaAs crystals [257].

1.5.2.1.1 Development of the Original RCA Wafer Cleaning Procedure

The development of this optimized wet-chemical cleaning procedure for Si wafers by Kern and Puotinen [101] proceeded concurrently with the contamination studies described in the previous section, which provided basic information on adsorption and desorption characteristics of many contaminants. In formulating a design plan, it was realized that the first step should remove the ever-present organic contaminants to expose the Si surface by use of a wet oxidant. Adsorbed ions and deposited trace metals would be removed by solubilizing with strong oxidant reagents. To prevent redeposition of the dissolved ionic contaminants, a complexing reagent would be required. The cleaning reaction must be selective so as not to corrode Si. The cleaning mixture to be formulated should consist of reagents that leave no residues and are readily available in pure form. Thermodynamic reasoning based on the oxidation potentials of several possible candidates were an important consideration in the reactant selection.

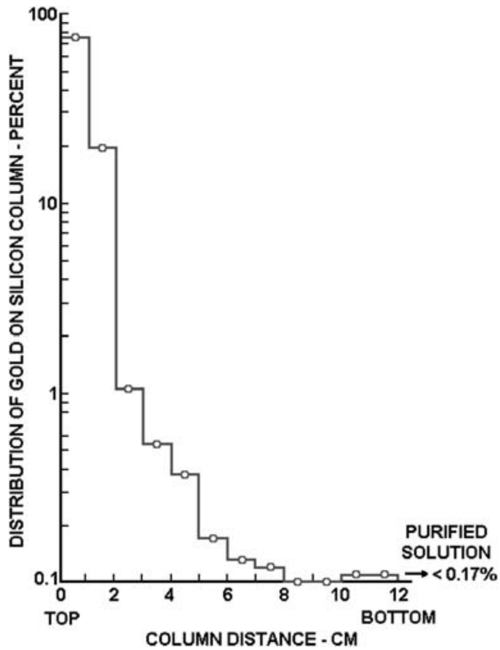

FIGURE 1.5-18 Distribution of Au deposits from a 49% HF solution containing Au ions with Au-198 as the radiotracer. The chromatographic column consisted of small Si crystal pieces [253]. *Used with permission from Semiconductor International.*

Fig. 1.5-19 reproduced from the original paper in 1970 shows the oxidation potentials of several common reactions as a function of pH. It can be seen that the oxidizing power increases with decreasing (more negative) electrode potential. For equivalent concentrations, the peroxide oxidation is the most powerful oxidizing reaction shown, indicating that H_2O_2 should be the primary reagent in the solution. The reactive additives selected were NH_4OH for the first solution, creating SC-1, at high pH for removing organics and for complexing some of the metals. Hydrochloric acid was selected for the second solution, creating SC-2, at low pH for removing remaining metals, hydroxides, and alkalis. DIW would be used as the diluent and a temperature would be determined for optimal reactivity. The exact formulations developed on this basis were described in detail in Sections 1.4.2.3 and 1.4.2.4. It was also mentioned that the poor stability of H_2O_2 in SC-1 and SC-2 requires the use of freshly prepared mixtures.

The graph presented in Fig. 1.5-20 dramatically illustrates this instability. Several improvements in the original RCA cleaning procedure were also described in this section [102–107], including the important use of megasonics [104].

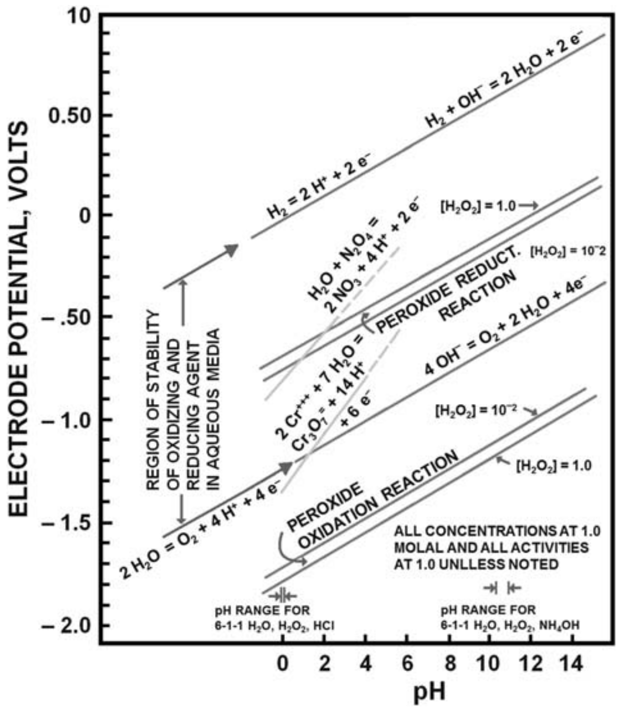

FIGURE 1.5-19 Electrode potentials versus pH for various redox systems at 25°C [101]. *Used with permission from RCA Review.*

1.5.3 Period from 1972 to 1989

1.5.3.1 Chronological Survey of the Literature on H_2O_2-Based Cleans

Beginning in 1972, independent investigators examined and verified by various analytical methods the effectiveness of the RCA cleaning method as published in 1970 [101]. Below we are reviewing work specifically on wafer cleaning pertaining to H_2O_2 solutions published during this period.

In 1972, Henderson presented an evaluation of SC-1/SC-2 cleaning by use of high-energy electron diffraction and Auger electron spectroscopy as analytical tools [258]. He concluded that the process is well suited for wafer cleaning before high-temperature treatments, as long as quartz-ware is used in the processing. A final etch in HF solution after SC-1/SC-2 caused C contamination and surface roughening during vacuum heating at 1100°C because of loss of the protective 1.5-nm-thick C-free oxide film remaining after SC-2. Meek et al. [259] investigated the removal of inorganic contaminants, including Cu and heavy metals, by several reagent solutions from silica-sol (very fine slurry) polished wafers. Using Rutherford backscattering, they

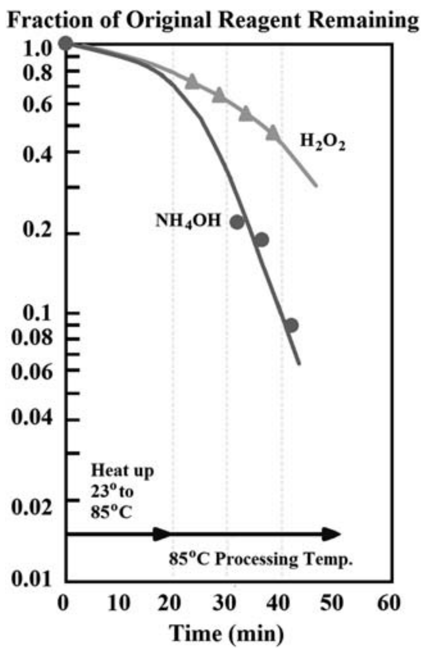

FIGURE 1.5-20 Decrease of the H_2O_2 and NH_4OH concentrations in 5:1:1 SC-1 as a function of Si wafer cleaning use time at excessively high temperature and long time periods in an open vessel. The decomposing H_2O_2 emits O_2, and the NH_4OH gives off NH_3 gas [103]. *Used with permission from Semiconductor International.*

concluded that SC-1/SC-2 preoxidation cleaning removes all elements heavier than Cl. Sulfur and Cl remained after SC-1, SC-2, or other cleaning procedures at the high concentration of E16/cm^2. SC-1/SC-2 cleaning eliminated Ca and Cu much more readily than did HF–HNO_3.

Amick [260] reported the presence of Cl on Si after SC-2 and of S after H_2SO_4– H_2O_2, using spark source mass spectrometric analysis. In 1976 Kern and Deckert published a brief review of surface contamination and semiconductor cleaning in a book chapter on chemical etching [85]. Murarka et al. [261] studied methods for oxidizing Si without generating stacking faults and concluded that SC-1/SC-2 before oxidation is essential for this purpose. Gluck [262] discussed removal of Au from Si by a variety of solutions. The desorption efficiency of SC-1 was more effective than that of SC-2, but the recommended sequential treatment of SC-1 followed by SC-2 was found to be the most effective method for removing Au at high surface concentrations (1×10^{14}/cm^2 range) [262].

Peters and Dekert [263] investigated photoresist stripping by solvents, chemical agents, and plasma ashing. SC-1 cleaning was the only acceptable technique, which completely removed the residues [263]. Burkman [108] reported on desorption of Au with several reagent solutions by centrifugal spraying. SC-1 was much more effective than H_2SO_4− H_2O_2, while the SC-2 alone showed poor efficiency [108], as expected.

Phillips et al. [264] applied SIMS to determine the relative quantities of contaminants on Si. Cleaned wafers were contaminated with gross quantities of numerous dissolved inorganic compounds and then cleaned by immersion or spray techniques using various reagents, including aqua regia, hot fuming HNO_3, and H_2SO_4−H_2O_2. The lowest residual concentrations were obtained by spray cleaning with H_2SO_4−H_2O_2 followed by the SC-1/HF/SC-2 sequence [264]. Goodman et al. [265] demonstrated by minority-carrier diffusion-length measurements the effectiveness of SC-1/SC-2 for desorbing trace metals on Si [265]. Kern [102] published a review of the RCA wafer cleaning procedure on the occasion of a Citation Classic declaration of the original 1970 paper [102].

In 1983 Watanabe et al. [266] reported dissolution rates of SiO_2 and Si_3N_4 films in SC-1. The rate of thermally grown SiO_2 films in SC-1 during 20 min at 80°C was a constant 0.4 nm/min, a significant rate for structures with thin oxide layers. The etch rate of CVD Si_3N_4 films was 0.2 nm/min for the same conditions, but measurements by Kern in 1981 (published in 1984) indicated much lower oxide dissolution rates under nearly identical conditions [103]. Film thicknesses were measured by ellipsometry after each of four consecutive treatments in fresh 5:1:1 SC-1 solutions at 85°C and totaled only 7.0 nm/80 min or 0.09 nm/min. Under the same conditions, 6:1:1 SC-2 solution showed no losses. Similar results averaging 0.13 nm/min were obtained with thermal SiO_2 films grown on lightly or heavily doped Si. Wafers from the same set were used to determine the etch rates of exposed Si in SC-1 solutions with decreasingly lower H_2O_2 content. Little etching or attack of Si occurred (less than 0.8 nm/min) even when the H_2O_2 concentration was reduced by 90% [103].

Bansal (1984, 1985) reported extensive results on particle removal from Si wafers by spray cleaning with SC-1/SC-2, H_2SO_2−H_2O_2, and HF solutions of various purity grades. He found the RCA cleaning solutions to be the most effective [267,268]. Shwartzman et al. [191] described simultaneous removal of particles and contaminant films by megasonic cleaning with SC-1 solutions. Ishizaka and Shiraki [269] showed that atomically clean Si surfaces for MBE can be prepared below 800°C in UHV by thermal desorption of a thin (0.5−0.8 nm) passivating oxide layer that protects from C contamination. The layer was formed by wet oxidation (HNO_3, SC-1) and HF-stripping, terminating with an SC-2 type treatment.

Wong and Klepner [270] used XPS analysis to examine Si after wet chemical treatments. RCA cleaning without buffered HF stripping resulted in about 30% of the Si atoms in the top 1.0 nm being oxidized, whereas with a final BHF step less than one monolayer of suboxide coverage resulted [270]. Grundner and Jacob [271] conducted extensive studies of Si surfaces after treatment with SC-1/SC-2 or 5% HF solutions, using X-ray photoelectron energy loss spectroscopy. Oxidizing solutions produced hydrophilic surfaces, whereas HF solutions led to hydrophobic surfaces consisting mainly of Si—H with some Si—CH_x and Si—F structures [271].

Becker et al. [272] reported on decontamination by using different reagent sequences. SIMS analysis was used to test for the removal of Na, K, Ca, Mg, Cr, Cu, Al, and particle impurities. The best cleaning sequence for metallics was H_2SO_4—H_2O_2/SC-1/HF/SC-2. Reversing the order of SC-1 and HF was most effective for particle removal and slightly less so for metal ions [272]. Kawado et al. [273] found by SIMS that Al on Si wafers originated from impure H_2O_2 used in SC-2. Very high concentrations resulted if Pyrex vessels were used in the processing instead of fused quartz [273]. In 1986 McGillivray et al. [274] investigated effects of reagent contaminants on MOS capacitors. Low field-breakdown was more prevalent if preoxidation cleaning with SC-2 was terminated with HF solution instead of omitting it. No other significant differences in electrical properties resulted from these two treatments.

Lampert [275] examined growth and properties of oxide films on Si in various solutions, including SC-1 and SC-2. Gould and Irene [276] studied the influence of preoxidation cleaning on Si oxidation kinetics. They found significant rate variations depending on treatments (SC-1/SC-2/HF, SC-1, SC-2, HF, no clean). Ruzyllo [277] reported on similar experiments and found that various preoxidation cleans seem to affect structure and/or composition of the subsequently grown oxide films. Slusser and MacDowell [21] found that sub-ppm levels of Al in H_2O_2 used for SC-1/SC-2 causes a substantial shift of up to 0.2 V in the flat band voltage of a dual dielectric. Aluminum concentrates on the wafer surface and the basic media such as SC-1 can lead to five times higher levels than acidic (SC-2) solutions [21]. In 1987 Kern and Schnable reviewed wafer cleaning processes in a book chapter on wet-chemical etching [86].

Probst et al. stated that for achieving predictable diffusion from implanted doped polySi into single-crystal Si, an SC-1/SC-2 treatment of the substrate before polySi deposition is imperative [278]. Khilnani [23] discussed various aspects of semiconductor cleaning, including the RCA process. Peterson [279] showed that the exact sequencing of cleaning solutions (H_2SO_4—H_2O_2, SC-1, SC-2, HF) could have dramatic effects on particle levels.

In 1989 Morita et al. [280] reported on Si surface contamination from SC-1/SC-2, finding traces of Na, Al, Cr, Fe, Ni, and Cu and reporting that the absence or presence of an SiO_x layer on the Si surface strongly affects adsorption. Desorption of Al and Fe deposits was most effective with HF—H_2O and that of Cu and Cr with SC-2. The same authors [281] postulated that metals of high enthalpy of oxide formation adsorb on the oxidized surface by oxide formation, whereas metals of high electron negativity deposit electrochemically on bare Si. Gould and Irene [282] studied the etching of native

SiO_x films and of Si in NH_4OH-H_2O, BHF, and SC-1 by ellipsometry. Severe Si surface roughness resulted from NH_4OH, less with BHF, and none with SC-1 [282].

Ohmi et al. [193] compared particle removal efficiencies of several cleaning solutions. They found that 5:1:1 SC-1 efficiently removes particles larger than 0.5 μm, but increased the density of those smaller than 0.5 μm (haze) unless the NH_4OH concentration was decreased to one half or less, in which case both types of particles were reduced efficiently [193]. Menon et al. [179] evaluated effects of solution chemistry (5:1:1, SC-1, DIW) and particle composition on megasonic cleaning efficiency at various power levels, concluding that megasonics can provide wafer cleanliness levels not previously attainable [179].

1.5.3.2 Other Important Advances

Important developments in other areas related to wet-cleaning of wafers should also be mentioned with representative key references. Significant advances in the physics of contaminant particles have led to a better understanding of adhesion, of submicron behavior in liquids and gases, and of their transfer mechanisms to solid surfaces. Most of this work, such as that by Bowling [283], Mishima et al. [195], and Menon et al. [179,284], was published between 1985 and 1989 and has resulted in improved high-purity processing and in more effective removal of particles from wafer surfaces, especially by the application of megasonics.

Advances in dry cleaning of semiconductor wafers are exemplified by the work of Vig [285], Kaneko et al. [286], and others who extended the use of UV/O_3 for removal of organic contaminants from semiconductor surfaces. Mishima et al. [194,195] and Ohmi et al. [193] published research on wafer drying techniques, especially on the subtleties involved in IPA vapor drying. The introduction of anhydrous gas etching and HF/H_2O vapor etching for removing oxide films from wafers to avoid particle contamination was pioneered by Claevelin [287], Clements et al. [288], and Duranko et al. [289]. This major advancement set the stage for vapor-phase cleaning technology. Early investigations were conducted with the then-novel remote plasma cleaning techniques by Fountain et al. [290].

Progress in another area associated with wafer cleaning concerns the refinement of microanalytical, chemical, and instrumental methods for detecting and quantifying trace contaminants and for exploring the atomic structure and morphology of semiconductor surfaces. Elucidating the chemical surface reactions of HF with Si and the resulting passivated, hydrogenated Si surface are additional examples. Papers on this subject were published by Burrows et al. [291], Hahn et al. [292], Zazzera and Moulder [293], and Chabal et al. [294].

1.5.4 Period from 1989 to 1992

This is the period of literal explosive growth of wafer cleaning science and technology with continually accelerating rate of progress. Rather than attempting a comprehensive coverage, we will confine the presentation to highlights achieved in (1) wet-chemical cleaning processes and (2) gas-phase cleaning methods. Much of this information is contained in the literature cited in the introduction, Section 1.2.1.

1.5.4.1 Wet-Chemical Cleaning Processes

New observations on the performance and effects of H_2O_2-based cleaning solutions led to some modifications of the RCA SC-1. In addition, high-purity chemicals had become available, including Al-free H_2O_2, low-particle HF, and low-metal HCl and NH_4OH solutions that led to much lower trace metal contamination.

Van den Meerakker and Van der Straaten [295] elucidated the mechanism of Si etching inhibition by H_2O_2 in SC-1 and the kinetics of etching. They reported a half-life for the standard 5:1:1 SC-1 of 16 min at 70°C and 9.3 min at 80°C. The authors stated that no etching occurred at 70°C as long the H_2O_2 concentration is at least 3×10^{-3} M to passivate the Si surface, which is 0.2% of the H_2O_2 concentration in 5:1:1 SC-1. In other words, no etching occurred as long as at least 1/500 of the H_2O_2 was present, which further extends our previous results [101].

Tanaka et al. [296] reported Si etch rates for 5:1:1 SC-1 of 0.5 nm/min at 75°C and of 0.8 nm/min at 85°C, with higher values for decreasing H_2O_2 or increasing NH_4OH concentrations. These values are higher than those determined by Kern, which averaged 0.05 nm/min at 80–85°C [103]. The discrepancy may be caused by the long etch time used by Tanaka et al. that could lead to a loss of the etch-protective oxide layer [295]; variations in the Si properties could also be the cause [103]. Although these etch rates are relatively low, one should not exceed the SC-1 treatment temperature of 70°C and the time of 5 min.

Microroughening of the Si surface as a result of nonuniform microetching by 5:1:1 SC-1 was investigated by many researchers, including Miyashita et al. [297], Ohmi et al. [298,299], and Heyns et al. [300]. This effect had become detectable only with the advent of atomic force microscopy and can have detrimental consequences on the breakdown voltage of thin gate oxide films, as reported by Meuris et al. [35], Verhaverbeke et al. [36,40], Ohmi et al. [299], and Heynes et al. [300]. Miyashita et al. [297] reported that a 10- to 100-fold reduction of the NH_4OH concentration in the original 5:1:1 H_2O:H_2O_2:NH_4OH SC-1 mixture eliminated roughening and enhanced the removal of particles, contrary to the results by Meuris et al. [35]. A reduction of the NH_4OH concentration to 5%–10% of that used in the conventional 5:1:1 SC-1 did not impair the desorption of Fe, Cu, Zn, and Ni from the Si surface. Meuris et al. [35] proposed a ratio of 5:1:0.25 as the best compromise for a modified SC-1 in terms of particle removal efficiency and avoidance of microroughening. The authors cited in this paragraph also examined the correlation between these effects, metal contamination, and electrical properties of grown oxide films [35,36,40,112,299,300].

Sakurai et al. [296] reported that the thickness of the chemically grown oxide films on Si during SC-1 cleaning does not depend on temperature, time, and solution composition (except for very low NH_4OH concentrations). The thickness of the films was 0.5–0.6 nm as determined by XPS analysis or 1.2 nm as measured by ellipsometry.

New results on wet-chemical cleaning by various processes and techniques were reported during this period by a number of researchers [28–40,111,112, 133,134,180,300–310] as follows: Heyns et al. [300] found that a dip of Si wafers in dHF after SC-1/SC-2 removes any metal contaminants that may still be present, without introducing new impurities. However, postcleaning exposure of the wafers to HF solutions will cause recontamination by organics, particles, and possibly trace metals. According to Rubloff [302], HF should be applied only in the case of subsequent low-temperature epitaxial vapor growth, where the absence of an oxide layer is crucial. Treatment by HF is not appropriate as preoxidation cleaning where the formation of a passivating oxide layer is essential to prevent thermal surface etching and roughening of the Si.

Verhaverbeke et al. [112] investigated the characteristics of HF-treated Si surfaces as a function of immersion time in dHF. They demonstrated the importance of forming a perfectly passivated surface, as evidenced by contact angle measurements, to reduce particle deposition. HF-last cleaning was more beneficial in terms of metallic contamination removal, as compared with RCA cleans. The improved processing led to superior oxide integrity [112]. Grundner et al. [303] investigated surface composition and morphology of Si wafers after dHF treatments by means of instrumental surface analytical and angle-resolved light-scattering techniques. Hirose et al. [304] studied the chemical stability and oxidation kinetics of H-terminated Si surfaces after treatments in HF and BHF solutions, and Chabal [305] explored H-termination, atomic structure, and overall morphology in detail.

Anttila and Tilli [111] showed that replacing SC-2 with very diluted mineral acids, e.g., $1:1 \times 10^4$ $HCl:H_2O$, can remove several metallic contaminants (and metal hydroxides) without introducing as many particles as SC-2 does. The benefits of SC-1, which leaves the surface free of particles and organics, are combined with the benefits of dilute acid at room temperature, which removes metals efficiently without adding particles. Kniffin et al. [306] showed that the type of chemical bonding of metallic impurities to the Si surface plays an important role in determining the cleaning efficiency of a wet-chemical processing sequence. Poliak et al. [307] compared the effects of various wet-chemical cleaning sequences for removing metallic contaminants.

Shimono [133] demonstrated that aqueous 1% H_2O_2–0.5% HF has a higher efficiency for removing metallic impurities than do conventional cleans; it also etches native oxide films and a surface layer of Si. Takizawa and Ohsawa [134] have used a similar approach by employing the classical HF–HNO_3–HF Si etchant in very dilute form (0.025%–0.1% HF in HNO_3), so that a Si etch rate of 3–60 nm/min results without a substantial etching of thermal SiO_2 patterns.

Hariri and Hockett [143] showed that replacing the NH_4OH in SC-1 with choline (trimethyl-2-hydroxyethyl ammonium hydroxide) plus a surfactant reduces oxidation-induced stacking faults in Si better than RCA cleans do and may improve removal of heavy metals. Lowell [308] used choline clean after a deglazing etch for doped polySi layers with dHF, generating a thin protective oxide film. Menon et al. [309] showed, however, megasonic/SC-1 to be more effective for removing particles than choline mixtures. Syverson et al. [310] conducted temperature optimization tests for megasonic particle removal in SC-1/SC-2, which revealed that 55°C is the most effective temperature. Major reduction in wafer particle densities was achieved by this treatment in an advanced manufacturing environment. Many other aspects of particle contamination and removal were covered in the volumes of Refs. [10,11].

Tong et al. [311] reported that ozonated DIW used with conventional aqueous chemicals has good cleaning efficiency; concentrations of residual metals and particles were found to be equal or lower than those with conventional RCA cleans. Matthews [72] carried the preparation of high-purity aqueous chemicals from gaseous precursors a step further by using NH_3 for preparing NH_4OH and HCl for HCl solutions, in addition to O_3 for H_2O_2. This method of reagent synthesis will undoubtedly become an important future technology.

An important area related to wafer cleaning concerns reagent recycling and repurifying. Davison et al. [69] described the reprocessing, properties, and applications of high-purity aqueous HF. Doshi et al. [312] state that impurity levels in this ion exchange–purified HF were routinely below 1 ppb for 34 elements and that it performed significantly better in the production of MOS memory devices than high-purity imported HF. Davison [313,314] and Hsu [70] reviewed the technology of reprocessing and ultrapurifying both H_2SO_4–H_2O_2 (piranha etch) and aqueous HF.

1.5.4.2 Vapor-Phase Cleaning Methods

A number of outstanding papers should be noted in this technology: The removal of native, grown, or deposited oxide films on Si was accomplished previously (1988) by use of HF gas or vapor etching [287–289]. The chemical mechanisms underlying these vapor-phase etching processes were elucidated by Helms and Deal [315]. Systems for implementation of these processes for oxide removal were reported during this period by many authors, e.g., Deal et al. [316], Ohmi et al. [317], Onishi et al. [318], Iscoff [319], Wong et al.

[320], Nobinger et al. [321], and Deal and Helms [322]. The addition of methanol instead of H_2O vapor to anhydrous HF minimizes the formation of the solid reaction products encountered with the HF−H_2O vapor etching systems, as observed by Izumi et al. [323].

Significant progress was made by the implementation of physical−chemical methods, such as glow discharge plasma reactions, for removing thin oxide films and certain contaminants from semiconductor wafers. Comfort [324] examined the thermodynamic parameters governing high-temperature thermal desorption of oxides and low-temperature removal of oxides for preepitaxial surface cleaning of Si. Reif [325] discussed in situ low-temperature cleaning for preepi Si growth; Lier [326] examined the impact of Si surface treatments before epitaxy and gate oxide growth; and Kalem et al. [327] reported on surface cleaning before the formation of SiO_2/Si interfaces. Tasch et al. [328] reviewed recent results on low-temperature in situ preepi cleaning of Si by remote plasma-excited hydrogen in ultrahigh vacuum. Hattangandy et al. [329] applied atomic H, dissociated from H_2 by remote noble gas discharge, for the low-temperature cleaning of Ge and GaAs surfaces. Frystak and Ruzyllo [330] used remote plasma cleaning as a preoxidation treatment for Si. Gas mixtures of O_2, HCl/Ar, and NF_3/H_2/Ar were used to remove organics, metallic impurities, and thin oxide films. Finally, Chang [331] described an in situ plasma cleaning and device passivation process for GaAs.

Removal of metallic contaminants is best accomplished by thermal, chemical, or photochemical vapor-phase reactions. Gluck [332] reported that nitric oxide can volatilize Cu from Si surfaces at 500°C and that Au reacts with a mixture of nitric oxide and HCl at 900−1000°C to form a volatilizable compound. The formation of volatile nitrosyl complex compounds with various metals and the thermal reaction with NO/HCl/N_2 may be promising approaches to vapor-phase metal removal.

Formation of volatile organometallic complexes with other reactants is also possible. Ivankovits et al. [333] reported that reacting Fe and Cu on Si with 1,1,1,5,5,5, hexafluoro-2,4-pentanedione followed by volatilization at 300°C can reduce their concentrations. These early results are promising, but a great deal more work is needed to develop predictable and efficient processes. Wong [334] reported results on a preoxidation treatment of Si wafers with HCl/HF vapor mixtures, which was effective in reducing the detrimental effects caused by traces of heavy metal contaminants; the oxide lifetime improved by 25%.

Low-temperature photochemical reactions have a great potential for transforming metallic contaminants into volatilizable compounds. UV radiation is usually employed as the radiation source with halogens as the reactants to generate highly reactive halogen radicals. Ito et al. [335,336] utilized highly purified chlorine radicals to reduce the surface concentrations of Fe, Mg, Al, and Cu to levels lower than those attainable with conventional wet cleans. Native oxide layers on Si can be etched off with fluorine radicals generated by the same photoactivation process [336].

The well-known UV/O_3 reaction for removing organics from surfaces, reviewed by Vig [337], was applied to the cleaning of GaAs in epitaxial deposition processes by Pearton et al. [338] and by Kopf et al. [339] for reducing defects. Bedge and Lamb [340] studied the kinetics of the process and reported on experiments and modeling.

The status of particle removal by promising dry-cleaning techniques was noted by several investigators. McDermott et al. [341] described an argon-aerosol jet technique where frozen particles of Ar are created and impinge at high velocity onto the wafer surface. Micron-size and smaller particles are dislodged kinetically by the collision and are entrained in the gas stream and removed from the system. Bok et al. [342] reported on the theory and practice of using supercritical fluids, such as supercritical CO_2, which can penetrate into deep IC structures and effect complete removal of particles and other contaminants. The supercritical liquid is first forced into trenches during compression in the pulsating pressure cycle. Subsequent expansion between supercritical and subcritical pressures dislodges particles and causes their ejection with tremendous force.

Removal of particles in a vacuum system compatible with a dry cleaning sequence is technologically very difficult. Particle detachment by electrostatic techniques is ineffective. A promising technique using a laser beam was demonstrated by Allen [343,344], who showed that pulsed-laser radiation is capable of removing particles from surfaces. A moisture film is condensed between the particles and the wafer surface and is then explosively evaporated by a laser beam of appropriately tuned wavelength. The dislodged particles are then swept out of the system with a jet of inert gas.

1.5.5 Period From 1993 to 2006

1.5.5.1 Trends and Milestones

This 13-year period from 1993 to 2006 was marked by tremendous changes and remarkable progress in the science and technology of Si wafer cleaning and surface conditioning. A number of major milestones can be identified in this era of intense research and development that have led to outstanding improvements in traditional processing methods and in the generation of entirely new ones. The progress has been gradual but rapid, reflecting the fast-moving industry of high-end microchip manufacturing with its stringent technical demands.

The driving forces underlying this development are the cleaning and conditioning requirements for semiconductor device structures featuring novel materials and continuously shrinking chip-level geometries. The efficient removal of microcontaminants from these ultrasensitive surfaces has become critically important and is the prime objective of all cleaning processes.

In the past, wafer cleaning and surface preparation were considered more of an art than a science. This view has definitely changed in recent years as processing and characterization methods have attained considerable scientific sophistication. The semiconductor industry now has a much better understanding of the chemical and physical mechanisms governing the transfer of contaminants to and their removal from Si wafers and other film surfaces, in part because of advances in instrumental ultratrace analysis and electrical methods of testing. Silicon surface concentrations of critical metallic impurities were reduced to levels below 10^8 atoms/cm^2, approaching the analytical detection limit; particle size measurement limits were reduced to below 50 nm.

The emergence in this era of new technologies for manufacturing advanced ICs has represented a new challenge to cleaning applications, especially in BEOL postpatterning cleaning of Cu interconnect lines and low-k interlayer dielectric films, which are among the most widely used new components. The cleaning of Cu surfaces demands special requirements because of the ease of oxidation. Dielectric films with lower dielectric constants than traditional SiO_2 are frequently of low density and require special cleaning treatments because of their poor chemical resistance.

Materials with high-k are another class of dielectrics that represent a special cleaning challenge because of their unique chemistry. These introduced materials are intended to alternatively replace the SiO_2 and SiON gate dielectric films in FEOL processes for several types of IC devices requiring high-k capacitors. Compounds of such materials may consist of the oxides of Hf, which is widely used, and Ta, Y, Gd, Sm, Dy, or Pr.

During this period single-wafer processing has become close to a mainstream technology as the wafer size has increased to 300 mm at the 90-nm, the 65-nm, and eventually the 45-nm process node. This transition from batch cleaning to single-wafer cleaning has also been combined with the development of economical spray tools and the introduction of new and improved chemistries. Organic contaminants have become recognized to be more critical in their effects than had been realized in the past, especially in respect to photoresist stripping and sidewall polymer residue removal. The older types of photoresist stripping agents based on hydrocarbon solvents, phenolic compounds, and amine active compounds have been replaced with *N*-methyl pyrrolidone containing a small amount of H_2O, and specialized formulations. Semiaqueous mixtures of hot hydroxylamine have been introduced for stripping photoresist over Al metallization.

Dry cleaning and surface conditioning technologies have experienced substantial progress, especially in the area of BEOL processing where wet cleaning has critical limitations. In addition, looking at R&D (research and development) conducted on the established processes based on HF-vapor etching, UV/O_3, UV/Cl_2, organochemical vapor-phase cleaning, and plasma stripping and cleaning, the most remarkable progress has been achieved with cryogenic aerosol and supercritical fluid cleaning. These cryogenic technologies have emerged from laboratory applications to IC production tools, especially cryogenic aerosol cleaning based on CO_2 and Ar/N_2 methods.

1.5.5.2 Liquid Processes and Wafer Drying Technology

It has become evident that wet-cleaning technology will remain the primary process for Si wafer cleaning and surface preparation in IC manufacturing for years to come, despite the early predictions and the availability of excellent and viable new dry-cleaning processes that have been developed during this period. An essential objective in wet-cleaning, next to the removal of contaminants, has been the reduction of processing chemicals, treatment time and DIW consumption by improved and optimized processes, simplified chemistry and advanced equipment design—all at lower defect levels.

The basic RCA standard cleaning sequence in its various modified forms still appears to be the major FEOL predeposition cleaning process in industry up to this time, although excellent alternative processes have been made available. Alternative FEOL cleaning solutions that have been described during this period include dHF solution in the "HF-last" step to produce a hydrophobic H-passivated Si surface [113−115]; variously diluted processing solutions used in "dilute chemistry" [110,116−120]; and O_3/DIW and ozonated chemical solutions as oxidants to replace H_2O_2 [100,121−132].

Other alternatives include microetching mixtures based on HF and an oxidant [115,121,133−138]; and Cu removal solutions containing HCl−HF−DIW [121,136−140]. Tetramethyl ammonium hydroxide−H_2O_2 mixtures have been used in spraying applications [145], and aqueous hydroxylamine solutions for postetch polymer residue removal [146,147]. A fast, cost-effective dilute-acid chemical mixture to perform postetch Al interconnect spray cleaning has been introduced recently [345].

The role of contamination introduced by contaminated wafers in immersion-type wet cleaning has been quantitatively assessed. It was shown that the quantity of metallic contaminants on the wafer surface after cleaning depends on the amount of metallic impurities introduced to the cleaning solution by the wafers initially [346].

Surfactants as additives to aqueous solutions have been extensively explored during these past 13 years to improve the overall cleaning efficiency [14,55,90,119,136,137,148−153]. The same is the case for chelating and complexing agents to enhance the cleaning activity of aqueous solutions and/or to prevent redeposition of contaminants [17,119,149,154−159]. Most of these chemicals have been available for many years, but have been evaluated and utilized only recently.

H_2O rinsing fundamental aspects have been studied in some detail [165,181−184], including physical/chemical mechanisms of patterned wafer rinsing [182−184] and optimization for single-wafer applications [182]. Acidification to prevent corrosion [186] and to avoid metal deposition on wafer surfaces has been proposed [52,165,185]. Megasonic rinsing for damage-free BEOL processing of Cu metallization has been described [187], and centrifugal spray rinsing [188] and rinsing in a closed system module

[189]. The cleaning effects of activated UPW have been found to be quite remarkable. This type of H_2O, so-called "functional H_2O," can be generated by ultrasonic irradiation of UPW that contains dissolved gases, generating H_2O_2 and reactive ionic species, such as OH^- radicals [346].

Variously operating commercial drying systems based on IPA have been developed [192]. It was realized that the purity of the solvent and its H_2O content are critically important to achieve an ultraclean and dry surface [193−195] free of the formation of watermarks and excessive IPA surface adsorption [347]. The "Rotagoni system," a single-wafer centrifuge operating at low speed and utilizing the Marangoni effect, was described for spin rinsing and IPA drying [182,199].

Complete new cleaning/rinsing/drying sequences that incorporate many of the processing steps and chemicals described in this section have been introduced during this period, mostly for FEOL predeposition applications. Most of these have been directed toward the use of single-wafer cleaning and are the subject of intense R&D, as already noted. The following new cleaning systems have been well documented: the Ohmi Clean [114,138,160−162], the IMEC Clean [52,113,163−165], the Diluted Dynamic Clean [166−168], and the Single-Wafer/Short-Cycle Clean [169−175,348] for BEOL cleaning. All these are directed toward the use of minimal quantities of chemicals, DIW, and processing time with superior results of cleaning and surface conditioning.

There has been an increased awareness or reducing H_2O consumption in wafer cleaning and rinsing applications. Great efforts have been made to conserve this precious and limited natural resource by optimized rinsing techniques [170,183,187,349,350], by introducing automated cleaning tools [351], by designing practical recycling systems [352,353] including electrodeionization [354], and, of course, by implementing the new cleaning systems listed in the previous section. Progress has also been made in the purification of H_2O, such as the introduction of dual, high-efficiency electrostatic filters to produce particle-free, UPW [355], to mention just one example.

1.5.5.3 Dry Cleaning and Surface Conditioning Processes

As previously stated, dry cleaning comprises the technologies of vapor-phase cleaning, plasma cleaning, and cryogenic cleaning.

Vapor-phase cleaning. Several papers have been published on vapor etching of native oxide films [201] and polymeric silicate residues and oxides in intermetal contact openings using anhydrous HF gas [202]. Additives of IPA, N_2, methanol, and H_2O_2 were used in several examples, including applications to MOS devices [203−208]. Etching studies of thermal SiO_2 with HF/H_2O vapor [209] and selective etching of native oxide films were reported [210].

Applications of UV/O_3 cleaning and surface conditioning were described in Ref. [211]. The UV-excited Cl_2 reaction was used under optimized conditions to remove trace metal contaminants in or on surface-damaged Si layers [205,209,212], including organics [213]. UV/Cl_2 pregate dielectric cleaning in a single-wafer reactor [214] and in a vacuum-clustered reactor has been described in Ref. [215].

Organochemical vapor-phase cleaning processes with chelates to form volatile metal coordination compounds have been reported in this period. A detailed study has been made on the removal of sub-monolayer quantities of numerous contaminant metals [220].

Plasma stripping and cleaning. Reactions involved in the removal of bulk photoresist with plasma-generated atomic O were described [221,222], including the use of downstream reactors to minimize ion-induced surface damage [223]. Water vapor addition can be beneficial [225,226] and so is the use of 10 vol% H_2−90 vol% N_2 in creating a nonoxidizing environment [201].

The removal of complex etching residues from deep Si trenches [227] using reactive chemical species in a downstream microwave discharge was reported [228]. Metal etch residues with corrosive Cl impurities [222] require the use of in situ H_2O-based downstream plasmas followed by wet-chemical cleaning [229]. Removal of etch residues by plasma treatments has been the subject of intensive studies to formulate effective processes. Predeposition cleaning and surface conditioning before gate dielectric formation under optimal conditions and plasma-assisted cleaning before epitaxy are additional critical operations that require carefully optimized procedures [231].

Cryogenic aerosol cleaning and conditioning. This emerging technology has received extensive attention during this period, as evident from the considerable amount of published papers. A theoretical analysis of cryogenic aerosol wafer cleaning has been reported [232]. Nondamaging cryogenic Ar/N_2 aerosols for cleaning IC device wafers with fragile interconnect structures have been used [233,234], including postpatterned BEOL cleaning of wafers with Cu metallization and low-k dielectrics [235]. Cryogenic aerosol processors for the fab are now available [233,234]. SCO_2 cleaning has been demonstrated to remove submicron particles and residues from post-CMP treated surfaces more effectively than is resulting from other cleaning processes [240,241]. Organic impurities appear to be removable by a liquid CO_2 phase [236−239]. Refined SCO_2 processing systems designed for wafer cleaning applications are known [237,239].

Supercritical carbon dioxide cleaning. As noted previously, supercritical carbon dioxide ($SCCO_2$) is capable of removing residues and submicron particle contaminants from IC device wafers with nanometer-size features without affecting porous low-k dielectric layers [237,238,242−244]. As a consequence, applications of $SCCO_2$ cleaning for IC manufacturing have received a great deal of attention, as evident from a number of listed references (including [242,245−247]); however this has not been embraced as a production-worthy technology.

1.6 MODERN CLEANING AND SURFACE CONDITIONING SCIENCE AND TECHNOLOGY

1.6.1 Trends and Milestones 2007 to 2017

The last 10 years have been marked with significant evolution of device structures and numerous additions of new materials into the IC device. With respect to wafer cleaning and surface conditioning, significant progress was made on tool development. Single-wafer cleaning has proliferated and is now considered mainstream. Wafer sizes of 300 mm are abundant and growing at a rate of 3–5 fabs/year. There are now one hundred 300-mm fabs many of them mega-fabs with greater than 100,000 wafer starts per month. Wafer sizes of 200 mm are also abundant and with the need for microcontrollers and other devices for the ample application, 200-mm fabs have made a come-back and now are on a growth spurt. Most of the fabs are logic and memory, but ASIC (application-specific IC), CMOS imager, analog, and power devices also are manufactured. In addition, foundry manufacturing supports the plentiful fabless design companies.

The cleaning and surface conditioning milestones during this period reflect the growing understanding of the effect of cleaning on higher yield. For example, edge management has become an important aspect of yield improvement; cleaning the edge of the wafer improves the yield of die at the outer 50 mm edge of a 300-mm wafer, which accounts for 55% of the die. Organic residue on the edge of the wafer will prevent adequate adhesion of W to Si, which will cause metal delamination. Optimized BARC (bottom anti-reflective coating) and resist removal plasma processing after etching, for edge management has been proposed to eliminate the film peeling [356].

Another milestone is the understanding of the cleaning and surface conditioning mechanisms and surface functionality. Silicon surface understanding is now being followed by understanding of Ge, compound semiconductor, and metal surfaces, such as W [357], Cu [358], Ru, and Co. Silicon surface understanding is further advancing with respect to chemical reactions, for example, the H-terminated surface from HF surface passivation [359].

Germanium is of particular interest because of its enhanced hole and electron mobility compared with Si. Germanium is a unique semiconductor material because its oxide is soluble in wafer and leads to cleaning issues. Particles can be removed relatively easily on Si with SC-1, but the same chemistry does not behave similarly on Ge; thus, novel chemistries that provide the same results on Ge as on Si are being researched [360]. Gaseous cleaning of Ge is proposed to prevent undesirable oxidation [361] when removing carbon contamination, which can form nucleation sites when depositing a film on the Ge surface or for enhanced oxidation. Oxidation in plasma after wet cleaning oxidizes carbon contamination and provides a surface with a controlled oxide layer [362]. Obtaining an oxide-free surface is a challenge for Ge. Chloride termination, in comparison with hydrogen termination for Si is the current desired functional state. Oxygen is known to cause Ge oxidation and for that matter Si oxidation; therefore, most processing tools have enclosed chambers under a blanket of N_2 to prevent unwanted oxidation. Additional challenge the industry is addressing is minimizing the surface roughness of Ge.

Future generations of devices will likely incorporate III-V semiconductor materials creating additional cleaning challenges. Besides well-known methods such as hydrogen plasma surface treatment [363], new passivation layers have been suggested for GaAs surfaces [364]. Understanding the oxidation of III-V materials is crucial for the development of optimized cleaning processes, which is critical for the interface management and also for surface conditioning during the wet cleaning and etching steps [365].

Other new materials in devices are requiring specialized cleaning, for example, power devices on silicon carbide wafer, which require the same surface condition as silicon wafer, $<$E10 atoms/cm^2 metal contamination, but are cleaned with different chemicals and methods [366].

Single-wafer cleaning is now mainstream; much development has been accomplished to provide a production-worthy critical cleaning process, high-temperature nitride removal, and damage-free particle removal. Learning includes prewetting of the wafer [367], prevention of dewetting of hydrophobic surfaces while the wafer is being processed [367], and prevention of bulls-eye patterns [368]. In addition, the physics of single-wafer cleaning has been published and much more work in this area is expected in the future, for example, the dynamics of liquid coverage on the wafer as it spins [369] and wafer charging induced by spinning [370].

The use of megasonic energy is falling for patterned wafers because of the extremely damaging effects on small device features. Understanding, characterization, and improvement of these processes may be able to extend the lifetime of megasonic energy for particle removal. For example, understanding the bubble growth and the control of the cavitation is crucial for reducing damage [371]. Although the use of megasonic energy may cause damage to patterned wafers, the technology is in use for post-CMP cleaning and blanket wafer cleaning. Improvement to wafer bonding have been shown with optimized single-wafer megasonic processing module that has a large contact area with the wafer to provide the needed uniformity [372]. Megasonic nozzles and systems are being understood with respect to frequency, energy, and gas content of the liquid to provide optimum cleaning efficiency [373–375].

Cleaning wafers in the manufacturing fab has been extended to wafer manufacturing, the need for cleaning before wafer bonding [376] for SOI formation for wafer manufacturing and 3D packaging is needed, especially on thin wafers. Thin wafer processing is being used for power devices, CMOS imagers, and high-density, stacked memory devices, in which cleaning is required on the thin wafers. Temporary bonding requires the cleaning of the adhesive of the back surface of a thinned wafer. Cleaning before bonding to achieve defect free surface is critical [377]. Handling of the thin wafer is also a challenge that must be addressed for wet single-wafer processing tools.

Cleaning the surface of particles before lithography has been an issue, especially back surface particles that can render a wafer out of focus with respect to exposure [378]. With the expectation that EUV will be included in select layers at the 7-nm node, cleaning particles off the front and back surface will be critical.

New formulations for cleaning include using self-assembled monolayers (SAM). Cleaning Al surface with SAM has been proposed by the researcher to prevent corrosion on the surface by providing an impenetrable barrier to O_2 [379]. SAM passivation has also been proposed for Ge, to control the surface states to prevent oxidation [380].

Other new formulations are based on electrochemical compatibility of new materials and the contaminants to be removed with the cleaning chemistry [381].

1.6.2 The Future of Cleaning and Surface Conditioning

As the 50th anniversary of Werner Kern's seminal paper [101] approaches in 2020, the semiconductor industry has grown from infancy to maturity. The science that was elusive in the past is much more understood thanks to the evolution of the analytical tools that were developed as a need to monitor the surface of the wafers. This will further assist in the understanding of the surface chemistry and physics in the future. Wet, plasma, and other cleaning tools will make further enhancements, not only to the process but to the tool itself; increasing throughput, managing contamination, and managing the surface state through a controlled environment.

Numerous consortiums and numerous academic institutes have research programs on cleaning and surface conditioning have produced significant papers in the past. This trend is hoped to continue as the IC industry introduces new materials, new device structures, and types of devices that challenge the cleaning and surface condition requirements and must develop new solutions to problems that have not yet been perceived.

1.7 SUMMARY AND CONCLUSION

This overview chapter has been presented in four major sections: (1) introduction, (2) wafer contamination aspects, (3) wafer cleaning technology, and (4) the evolution of wafer cleaning.

In the introductory section we briefly discussed the importance of clean and conditioned Si wafer surfaces in the manufacturing of microelectronic devices. Ultraclean Si and IC device surfaces that are essentially free of chemical contaminants and particulate impurities have been shown to be an absolute prerequisite for the economical fabrication of devices with high performance and reliability.

Damage-free and effective techniques for cleaning and surface conditioning Si wafers before and after the various processing steps are now critically important because of the extreme sensitivity of the semiconductor surface and the often fragile nanometer-sized device features. The introduction of Cu interconnect metallization and new dielectric materials has necessitated the development of optimized BEOL cleaning procedures that are more compatible with these materials than conventional processes can provide.

The results of intensive R&D efforts in recent years have been reported in symposia and numerous publications of technical and scientific professional societies aimed at meeting the stringent demands of the integrated circuit manufacturing community. Of the 500–800 process steps required to produce an advanced type of IC, approximately 15%–20% of them are for cleaning, and up to 70% of these are based on aqueous chemistry. We have seen that vapor phase or dry cleaning processes are used mainly for cleaning device wafers in the later stages of production (BEOL) where special materials may not tolerate conventional wet-chemical procedures.

The section on wafer contamination aspects summarized the types and origins of contaminants and defectivity to provide a better understanding for the need of surface cleaning. Contaminants were grouped in four categories: (1) molecular films, mostly of organic nature, (2) ionic contaminants, (3) metallic impurities, and (4) particles of all sorts. Each category has its specific features of deleterious effects on Si device performance, which were briefly described. The key notion on contamination and defectivity should be *prevention*, which is less difficult to exercise than subsequent contaminant *removal*. The purpose of this section was to provide an introduction to Chapter 2 on contamination and defectivity.

The third and main section was intended as an overview of wafer cleaning and surface conditioning technology. Wet-chemical cleaning processes that were discussed include solutions based on HF and on H_2SO_4–H_2O_2 mixtures. The original RCA cleaning process and its development and subsequent modifications were described in detail because of their importance and widespread use. Alternative cleaning solutions were discussed also, including "HF-last," dilute chemistry applications, ozonated solutions, microetching mixtures, and several other solutions. The steadily increasing use of surfactants and chelating agents as additives was noted. Alternative wet-chemical cleaning/surface conditioning systems and their key features that were described include Ohmi Clean, IMEC Clean, DDC, and Single-Wafer/Short-Cycle Clean. Equipment and techniques for implementing wet-chemical cleaning, wafer rinsing, drying, and storing were addressed. The trend toward high-speed, single-wafer spray-processing was pointed out for use with 300-mm wafers.

The overview of dry-cleaning and surface conditioning processes included HF vapor etching, UV/O_3 cleaning for organic removal, UV/Cl_2 vapor-phase reactions for metal removal, and organochemical vapor-phase cleaning reactions to be described in more detail in Chapter 5.

Plasma stripping and cleaning processes were briefly discussed as an introduction to the plasma cleaning in Chapter 6, including bulk photoresist stripping, etch residue removal, and predeposition/surface conditioning treatments.

The emerging technologies of cryogenic aerosol and supercritical fluid cleaning were described as a brief introduction to Chapter 7, including applications of SCO_2 for cleaning and particle removal of sensitive IC structures.

Section 1.5.1.4 reviewed the chronological evolution of wafer cleaning science and technology. Detailed references with author identifications were provided for work published from 1950 to 1992.

Graphical examples were shown from a series of basic papers published by the author between 1963 and 1972 using radioactive tracer isotopes to investigate the nature and properties of surface contaminants. The concurrent development of the RCA Si wafer cleaning procedure was also described in this section.

The period from 1989 to 1992 reviewed important developments of wet-chemical cleaning processes and the introduction of dilute chemistry and ozonated solutions to replace H_2O_2. Vapor-phase cleaning methods based on HF, organometallics, glow discharge plasma, and photochemical reactions were described.

The 13-year period from 1993 to 2006 is reviewed in terms of trends and major milestones. Significant progress in science and technology of Si wafer cleaning and surface conditioning was achieved with remarkable sophistication and scientific insight. The cleaning of BEOL Cu interconnect lines with low-k dielectric interlayers by novel liquid and dry-chemical methods exemplified some of the typical changes. Refinements in liquid processes and wafer drying technology were described, including new cleaning/rinsing/drying sequences and systems for single-wafer FEOL and BEOL processing. Emerging BEOL dry-cleaning and surface conditioning technologies based on activated gases, plasmas, cryogenic aerosols, and $scCO_2$ for patterned IC device wafers were noted among recent advances.

The most recent period from 2007 to 2017 held significant progress in understanding the surface phenomena associated with cleaning. The advent of new materials has challenged the cleaning community, with the outcome being improvement in the processing tool, process components, and processes.

In conclusion, the cleaning and surface conditioning of Si wafers and device structures designed for the 20-nm to 10-nm (and eventually 7-nm and 5-nm) technology node has become a major cornerstone in the fabrication process of advanced Si ULSI microcircuits, and soon nanocircuits. Various cleaning processes—wet and dry—will likely perform jointly as a team in FEOL and BEOL processing to utilize the best features of each for any specific application. However, liquid-based cleans will most likely continue to be needed for the removal of particles and solid residues. Continuous R&D will undoubtedly meet the stringent future challenges of cleaning devices featuring new structural materials using innovative approaches and refined systems and equipment.

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